Implications of Climate Change for Water Resources in the Kunduz River Basin, Afghanistan

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ABSTRACT

Afghanistan is ranked as one of the poorest countries in the world, listed second from the bottom of the Human Development Index. Twenty-five years of conflict have increased people's poverty, while recurrent periods of drought have made the Afghan environment ever more fragile. Food insecurity is one of the biggest challenges facing Afghanistan where cereal production has not been sufficient to meet consumption requirements since 1976.

Agricultural production is heavily dependant upon seasonal snowmelt originating in the Hindu Kush. Climate change is expected to have severe consequences for these crucial resources.

The use of satellite sensors provides a suitable platform to monitor such environmental variability at high spatial and temporal resolutions. This is especially useful in countries such as Afghanistan were widespread lack of security restrict the collection of field data.

This project has used satellite data together with meteorological observations and hydrological data to assess historical trends in snow cover in the Kunduz river Basin in north-east Afghanistan. It also provides an analysis of the possible implications of future climate change on these water resources.

It was found that warming trends in both winter and spring, have resulted in reduced snow cover in the upper watershed regions of the Kunduz River Basin. This reduction has been detected over the past decade in all months (December – May) but is greatest at the end of spring, therefore signalling a contraction of the snow season. This trend is likely to continue with projected warming trends in the 21st Century altering the quantity and timing of surface flows.

Reduced snowcover together with earlier spring melt is likely to have serious implications for irrigated agriculture which is the main economic basis of the Kunduz River Basin. The frequency of spring floods and summer droughts is likely to increase as a result of more intense precipitation events combined with earlier, more rapid snowmelt and higher evaporation rates; all being driven by increasing temperatures.

These natural disasters place heavy tolls on livelihoods and communities in the KRB and are likely to continue to do so with increasing frequency.

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Abbreviations

ANDS – Afghanistan National Development Strategy
DJF – December/ January/ February
FAO – Food and Agriculture Organisation
GCMs - General Circulation Models
IPCC – Intergovernmental Panel on Climate Change
JJA – June/ July/ August
KRB – Kunduz River Basin
Masl – metres above sea level
MAM- March /April/ May
RCMs - Regional Circulation Models
SON – September/ October/ November
UNEP – United Nations Environment Programme

1. INTRODUCTION

1.1 BACKGROUND & RATIONALE

Afghanistan is ranked as one of the poorest countries in the world, listed second from the bottom of the Human Development Index. Twenty-five years of conflict have increased people's poverty, while recurrent periods of drought have made the Afghan environment ever more fragile. Food insecurity is one of the biggest challenges facing Afghanistan where cereal production has not been sufficient to meet consumption requirements since 1976. Afghanistan is therefore still heavily reliant on food aid. (ANDS 2005).

Afghanistan is an arid/semi-arid country and eighty per cent of it's water resources originate in the Hindu Kush mountains at altitudes over 2000m. These resources include permanent snowfields and glaciers as well as seasonal snowfall which accumulates during the winter months and melts during spring and summer, thus supporting perennial flow of all major rivers. Afghanistan is therefore heavily reliant on irrigation during the summer months, when there is virtually no precipitation, in the lowlands to support agricultural production.

These cyrospheric elements are highly susceptible to long term climatic change as well as inter-annual climatic variation. The ability to monitor both the extent of snow cover and the timing of depletion is invaluable in predicting flood frequency and intensity and to predict expected summer drought before the winter rains. The use of satellite sensors and interpretation of the data using remote Sensing tools provides a suitable platform to monitor such environmental variability at high spatial and temporal resolutions. This is especially useful in countries such as Afghanistan were widespread lack of security restrict the collection of field data.

The vulnerability of Afghanistan's water resources was demonstrated at a national level with disastrous consequences over the period 1999-2001, when below average precipitation levels combined with mild winters resulted in significant decreases in snowfall. Moreover, increases in the daily minimum temperatures caused early and accelerated snow melt in the Hindu Kush. This resulted in a substantial decrease in available water over the summer months for crucial irrigated crops (FAO 2004).

This trend is expected to continue. In the course of the 21st century, water supplies stored in glaciers and snow cover are projected to decline, reducing water availability in regions supplied by melt water from major mountain ranges, where more than one-sixth of the worlds population currently lives (IPCC 2007).

In Afghanistan, as other poor nations, both climate change and lack of adaptive capacity due to poverty pose very serious challenges to livelihoods. Predicted climate change scenarios anticipate winter temperature increases of approximately 4 degrees Celsius by the latter half of the 21st century. These

scenarios would have severe implications for snow pack accumulation and timing of snow melt and the irrigated agriculture that it supports.

In the capital of Afghanistan there is a saying: *May Kabul be without gold rather than without snow.*

1.4. AIMS & OBJECTIVES

Due to the diverse nature of the climatology of Afghanistan there is a real need for regional analysis of the impact of climate forcing on water resources. This is especially true in mountainous regions were General Circulation Models (GCMs) and even Regional Circulation Models (RCMs) typically perform poorly due to the climatic complexity of these areas.

Added to this fact is a real paucity of current and historical data relating to water resources in Afghanistan due to the political turmoil the country has experienced in recent decades

This project aims to address this lack of data and by providing an analysis of observed trends in snow cover in the Kunduz River Basin coupled with long-term meteorological data for the region. Additionally, it will use available climatic projections for the region to provide an analysis of climatic vulnerabilities that exist in respect to water resources.

Specific objectives

1. Investigate observed trends in historical snow cover in the upper watershed regions of the Kunduz River Basin using satellite data.
2. Interpret this snow data using meteorological datasets from the Climate Research Unit, UK
3. Analyse future climate projections (Hadley Centre GCM) for Afghanistan in terms of impacts on water resources and livelihoods in the Kunduz River Basin.

This project was undertaken in collaboration with Mercy Corps UK, an International humanitarian aid and development organization that has been working in Afghanistan for over 20 years and Landell Mills a development consultancy who are providing technical assistance to the Government of Afghanistan. This research will contribute to the Kunduz River Basin Programme (KRBP) a pilot for the concept of River Basin Management in Afghanistan which is funded by the European Commission and implemented by the Government of Afghanistan

2. BACKGROUND LITERATURE

2.1. Mountain Hydrology

Although mountain regions occupy a relatively small area in most river basins they play a critical and distinctive role in regional hydrological cycles. Mountains act as barriers to the flow of moisture bearing winds and control precipitation in neighbouring areas (Iyngararasan *et al.* 2004). Worldwide, mountain regions display a great deal of heterogeneity; however, they all share the combined central attributes of elevated altitude and relief, which together influence mountain hydrology extensively.

Elements which are strongly forced by altitude include temperature, precipitation, soils and vegetation. Reduced temperatures at higher elevations in mountainous regions means that less evaporation occurs and that precipitation is more likely to fall as snow. Additionally, precipitation is generally more frequent with greater volume than lowland areas. Reduced vegetative cover and shallow or non-existent soil profiles decrease the processes of transpiration and infiltration, thus increasing runoff (Weingartner *et al.* 2003).

Elements connected to relief include slope and aspect. Slope influences all hydrological processes through the effects of gravity, including important runoff generation and erosion. Steep topography concentrates moisture availability and produces significant fluvial discharges. Aspect creates a great deal of micro-climatic variation within a given altitudinal band. Incident radiation, snow melt, evaporation rates, precipitation levels, and erosive forces are all modified by aspect (Weingartner *et al.* 2003).

All these elements combine to make mountain areas effective collectors, managers and distributors of water with often distinct seasonal or diurnal rhythms.

Snow pack accumulation in mountain areas during winter represents an important water resource for many regions in the world. Seasonally stored snow, together with melt water from glaciers, produce discharges which feed rivers and replenishes groundwater resources during the summer months when demand is often highest due to increased evapotranspiration rates (Barnett *et al.* 2005). All of the world's major rivers originate in mountains and between a third and a half of all freshwater flows originate in mountain regions (Iyngararasan *et al.* 2004). Alpine and high-mountain areas therefore act as effective water towers for more populated lowlands.

The term snowmelt dominated basin has been used to signify a high degree of dependence on snowmelt within the hydrology of a region. Within these areas snow is a crucial component of the hydrological cycle and plays an important role in replenishing groundwater resources and maintaining perennial flow of rivers (Barnett *et al.* 2005). .

The Kunduz River Basin (KRB) is amongst the most critical examples of snowmelt dominated basin due to a continental climate with negligible summer precipitation; therefore snowmelt is often the only available input to streams and groundwater. Snowmelt dominated regions tend to lie at latitudes greater than 45 degrees (north and south) together with mountainous regions (such as the KRB) outside these latitudes; they support 1 billion people around the globe (Barnett et al. 2005).

2.2. Impacts of Climatic Change on Mountain Hydrology

Increases in the concentration of carbon dioxide and other greenhouse gases in the atmosphere has resulted in an average global surface air temperature increase of 0.76°C since the beginning of the twentieth century (Trenberth *et al.* 2007). Additionally, General Circulation Models (GCMs) predict a further average warming of 1.8C – 4.0C by the end of the 21st century (IPCC 2007).

There is a great deal of spatial variation in observed and projected effects of climate change at the global level. However, few climate model simulations have attempted to directly address the issue of climate change in mountain areas as the resolution of General Circulation Models (GCMs) are generally too coarse to allow for the topographic detail of these areas (Christensen *et al*. 2007). Nonetheless, mountainous regions are particularly vulnerable to the impacts of climate change (Beniston 2003, Barry 1990). This is largely a result of a high natural variation in mountain climates due to large altitudinal gradients and the rapid shifts in temperature and precipitation over short distances that this creates (Gerrard 1990). This makes mountain areas more sensitive to even small degrees of climate change. For example, permafrost often exists at temperatures close to melting point. Therefore, even small increases in temperature combined with the loss of cooling snow cover can lead to widespread melt and increased slope instability (Harris et al. 2003, Anisimov & Nelson 1996). Additionally, in temperate mountain regions, the snowpack is often close to its melting point, especially in spring, so that it may respond rapidly to minor changes in temperature (Iyngararasan *et al.* 2004).

Added to this inherent vulnerability is the fact that warming trends demonstrate a significant elevation dependency; temperatures will rise more in high mountains than at low altitudes (Beniston and Rebetez 1996). This is caused by depletion of snow pack and glaciated area at high altitudes with existing warming trends causing a feedback through reduced snow-albedo effect (Giorgi et al. 1997, Immerzeel 2007).

Probably the most important climatic sensitivity mountain regions exhibit is in the role they play in supplying water to downstream areas. As mountain regions are the source of 50% of global river systems the impact of climate change on mountain hydrology will have a significant effect on downstream users in populated lowlands (Arnell 1999). More specifically, one billion people depend on water supplied by snow melt dominated mountain basins (Barnett *et al*. 2005) and changes in hydrology, water availability and quality as a result of climate change are expected to effect these regions the most (Nijssen *et al.* 2001).

With increasing temperatures many areas will see an increase in proportion of precipitation falling as rain in winter leading to a decrease in snowpack accumulation together with a contraction of the snow season (Dankers and Christenson 2005, Stewart *et al*. 2004). These changes are likely to lead to shifts in the timing and magnitude of runoff and river basin discharge leading to significant changes in the temporal distribution of water availability (Immerzeel 2007).

Average winter runoff is likely to increase, as is the rate and intensity of snowmelt in spring causing earlier peak discharge thus reducing water availability and soil moisture in the summer months during peak demand (Ragab and Prudhomme 2002). These changes are likely to have profound effects on water management systems such as reservoirs and irrigation schemes (Ragab and Prudhomme 2002).

Warming of the atmosphere due to climate change means its moisture capacity is increased as is the process of evaporation. This leads to an intensification of the hydrological cycle with more frequent and heavier precipitation events likely (Huntington 2006). These shifts in intra-annual precipitation regimes and timing of runoff could result in greater flood or drought episodes depending on the season (Graham *et al*. 2007). In some regions there is more concern associated with flooding. Projected increases in precipitation levels particularly when coupled with intensified snowmelt rates are likely to have serious consequences in the form of increased flood frequency for many mountain regions (Aizen *et al*. 2006, Barnett *et al*. 2005, Singh & Bengtsson 2003, Kundzewicz *et al*. 2007). Additionally, it is worth noting that drought and flood can perfectly reasonable exist side by side within any given region, particularly in distinctly continental climates were precipitation is largely a winter/spring phenomenon.

Variation of winter precipitation is a crucial component of any analysis of changing hydrological regime in snow-melt dominated regions. While all models predict atmospheric warming with elevated greenhouse gas levels, there is still a great deal of uncertainty in predicting changes in precipitation patterns due to a greater spatial and seasonal variability than temperature change (Trenberth *et al.* 2007, Chen *et al*. 2002). In addition large scale climatic forcing by natural events such as ENSO can lead to anomalous precipitation patterns on scales ranging from seasons to decades (Beniston and Jungo, 2001). Overall, efforts at mapping historical trends in precipitation have been characterised by a large degree of error and discrepancy between models (e.g. Mitchell and Jones 2005, Chen *et al*. 2002, Trenberth *et al*. 2007) which highlights the difficulties of monitoring precipitation (Trenberth *et al.* 2007).

Basin level responses to climate change

The responses of the hydrological regimes of river basins to climate change depend on numerous factors such as the source of runoff, climatic conditions, physical characteristics of the basin and magnitude of projected climate change scenarios (Singh & Bengtsson 2004). River basins can be broadly grouped into

three types based upon the source of runoff: (1) rainfed basins – runoff is generated entirely from rainfall; (2) snowfed basins – runoff is generated from rainfall throughout the year but supported by winter snowfall which generally melts in its entirety over the summer months; and (3) glacier fed basins – runoff can be generated by all processes in (1) and (2) but additionally permanent snowfields and glaciers contribute to runoff (Singh and Kumar 1997).

Projected warming in the majority of mountain regions will therefore have differing implications for these three types of basin. In terms of the cryosphere, the immediate impact of warming tends to be opposing depending on whether these water resources are seasonal (seasonal snow) or perennial (permanent snowfields and glaciers),

An entirely snow melt fed basin will see a decrease in the proportion of precipitation falling as snow as the snowline rises with increasing temperature. This will lead to greater direct runoff during winter months but a reduced runoff in summer months due to a reduced snow pack stock supplementing rainfall (Singh and Kumar 1997).

However a glacier fed basin will experience increased runoff during summer months as melt rates increase with elevated temperatures – until of course the capital resource of the glacier has reduced to such an extent that increased melt rates do not compensate for the reduced ice volume (Singh and Kumar 1997, De Jong 2007).

Because of the complexity and variability of interactions between climate, topography, landcover and soils that exist in different regions there is a real need for local modelling at the basin level in order to predict the consequences of climatic change with a high degree of confidence.

Observations of climatic forcing on Snow cover

Satellite records dating back to 1966 indicate that mean monthly snow cover in the Northern Hemisphere has decreased at a rate of 1.3 per cent per decade during the last 40 years with greatest losses during spring and summer months (Lemke *et al.* 2007, Dye 2002, Brown 2000). This trend is explained by a highly significant correlation between spring surface air temperatures and spring snow cover (Dore 2005). Although winter warming has occurred in most regions winter snow cover has not decreased substantially. Additionally, longer regional time series based on station records and reconstructions suggest that Northern Hemisphere spring and summer snow cover in the past decade has been at its lowest extent in the past 100 years. (Brown 2000). This decrease has led to a shift in month of maximum SCA from February to January and a significant trend towards earlier snow cover disappearance in spring of 3-5days/decade over the period 1972 – 2000 (Dye 2002).

However in winter season and altitudes were temperatures are significantly below zero to buffer current warming trends, increases in snow cover have been observed in some areas, associated with increased precipitation levels. For example, the Tibetan plateau has seen an increasing trend in snow cover as current winter temperatures are sufficiently low enough for regional warming trends to be secondary to increased precipitation (snowfall) trends (Dahe et al. 2006, Zhang and Wang 2004).

2.3. Climate Change and Water Resources of Afghanistan

Afghanistan has a dry continental climate and 90% of the countries annual precipitation occurs between December and April, resulting from eastward circulating cyclones from the Mediterranean region known as Western Disturbances (Syed *et al.* 2006). The vast majority of this precipitation is intercepted by the presence of the Hindu Kush Mountains leaving the lowland plains largely dry and barren. Due to seasonality and altitude, precipitation tends to occur as snow in the Hindu Hush. There is very little precipitation in the summer months. Precipitation is well correlated with altitude, ranging from <100mm/year at altitudes below 1000m to >1000mm/yr at altitudes over 4000m. On average snowmelt contributes 150, 000 million m3 to the countries water resources and rainfall 30 000 million m3 (Uhl 2006). Snowfall in the Hindu Kush Mountains represents 83% of annual precipitation and therefore plays a crucial role in Afghanistan's hydrological cycle by supporting perennial flow of all major rivers. Both timing and quantity of snowmelt are important factors in the water management and agriculture of the region (Syed *et al.* 2006).

Predicating the extent and effect of climatic forcing on snow cover in the Hindu Kush is therefore of utmost importance in determining possible impacts on the livelihoods of the people of Afghanistan. However, the Hindu Kush not only exhibit difficulties of modeling as other mountain ranges but suffer from a lack of reliable meteorological data due to inaccessibility and political turmoil of recent decades. While detailed studies exist for many regions in the Central Asian mountain ranges of the Pamirs and Tien Shan (Marchenko *et al*. 2007 Aizen *et al.* 1997) and the Himalaya (Singh and Bengtsson 2007, Immerzeel 2007, Jianchu *et al*. 2007), very few exist for the Hindu Kush of Afghanistan.

Additionally, comparisons with other studies are problematic as the Afghan Hindu Kush lie at the nexus of the largely monsoon dominated Himalaya and the continental Central Asian Massifs. While many studies consider climate change in the context of the Himalaya Hindu Kush region (e.g. Barnett *et al*. 2005, Winiger *et al*. 2005), this is slightly misleading in terms of Afghanistan. Firstly, the Himalaya is dominated by the South Asian Monsoon with high levels of precipitation in the summer time and generally drier spring months (Chalise *et al*. 2003, Gerrard 1990). This is the converse of the Afghan Hindu Kush which experiences 90% of precipitation December to April and virtually none during the summer. Afghanistan is therefore much more heavily reliant on snow melt for summer water resources. Secondly, the Himalaya has vast glacial systems which provide summer runoff. Afghanistan is generally less reliant on glaciers but

much more dependant on seasonal snow pack (UNEP 2003). As discussed previously, the significance of this lies in the fact that increased temperatures will decrease the seasonal snow pack reservoir without the vast perennial ice stores the Himalayan glaciers possess to supplement summer river flows.

Climate Change Projections

By the end of the 21st Century the Central Asian countries of which, climatically, Afghanistan is part of, are projected to experience temperature increase of 3.7ºC which is well above the global mean of 2.5ºC (Christensen *et al.* 2007, MMD A1B model). Data from the HadCM3 GCM (UK) puts this figure even higher for Afghanistan itself at 3.55ºC - 4.5ºC (Mitchell *et al*. 2002).

Precipitation is less certain with projections for Afghanistan ranging from -1.3 to +14.1% in winter and -25.2 to -5.8% in summer (Mitchell *et al*. 2002). Although the summer precipitation decrease appears fairly significant, it will have minimal impact as very little precipitation occurs in summer. A larger effect on summer water resources will be felt by increased evapotranspiration rates with elevated temperatures. Increasing winter precipitation will enhance snow pack accumulation but this is balanced against a contracted altitudinal and seasonal extent of snow cover with increased temperatures.

Apart from variation in mean values of climatic variables, climate change is likely to have an effect on extreme events. Afghanistan has been found to be heavily influenced by both ENSO and NAO through a remote forcing mechanism which strengthens (weakens) the precipitation bearing Western Disturbance with the warm (cold) phase of ENSO and the positive (negative) phase of NAO (Syed 2005, Argawala 2001). A good example of this is the 1999-2001 drought which was the worst in the 50 year record across the whole of Central and South-West Asia (Argawala 2001). This was a combination of a protracted cold phase ENSO event together with anomalously high surface temperatures. Projections as to what extent remote forcing mechanism will be affected by warming remains uncertain due to the complexity of interacting climatic systems that must be accounted for in any modeling exercise (Trenberth *et al*. 2007, Tsonis *et al*. 2003).

3. THE KUNDUZ RIVER BASIN

3.1. Background

The Kunduz River Basin (KRB) covers a total area of 35,000 square kilometres and drains a north-eastern part of Afghanistan in to the great Central Asian Amu Darya (Oxus) River. The KRB comprises of a part of Bamyan province, most of Baghlan and Kunduz and the southern half of Takhar province. The basin forms part of the northern plains, bordering Tajikistan to the north and defined by the central Hindu Kush in the south. The altitudinal gradient is large and ranges from less than 500masl on the plains to over 5000masl in the Hindu Kush range; 44% of the basin is at an elevation greater than 2500masl (fig. 1, Riedmann and Beekma 2005).

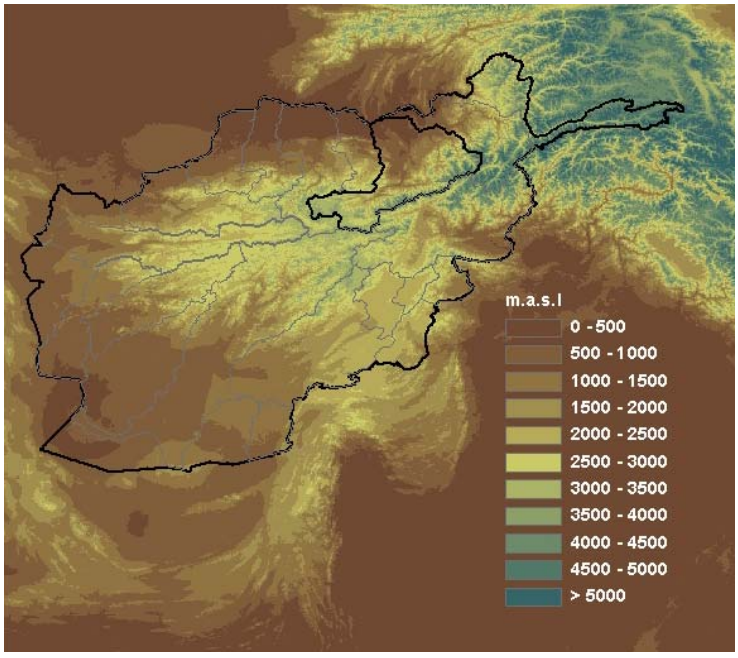


Fig 1: Topography of Afghanistan, Kunduz River Basin outlined in north-east.

The basin has a continental climate with dry hot summers and cold, moister winters (fig 2). Precipitation varies to a large extent with altitude and season, generally increasing with altitude and the majority falling during winter and spring, typically as seasonal snow. Annual precipitation varies from 350-500mm on the plains to 900-1100 in the upper watershed (fig. 3).

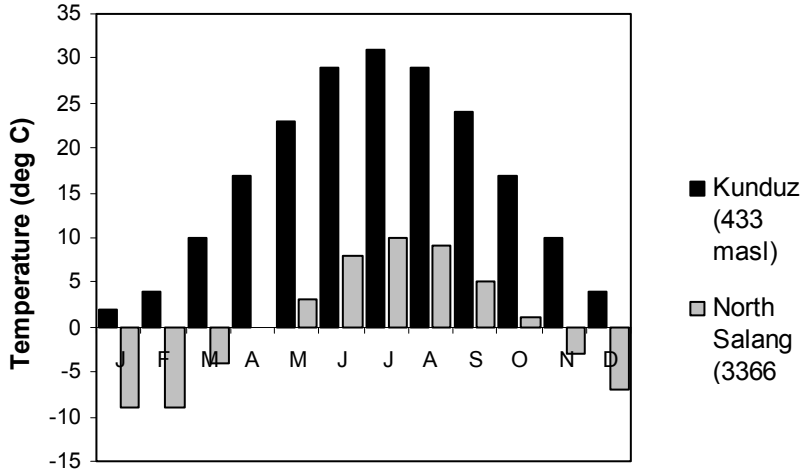


Fig 2: Mean monthly temperature in the KRB. Variation with altitude: Lowland plains = Kunduz, Upper watershed = North Salang.

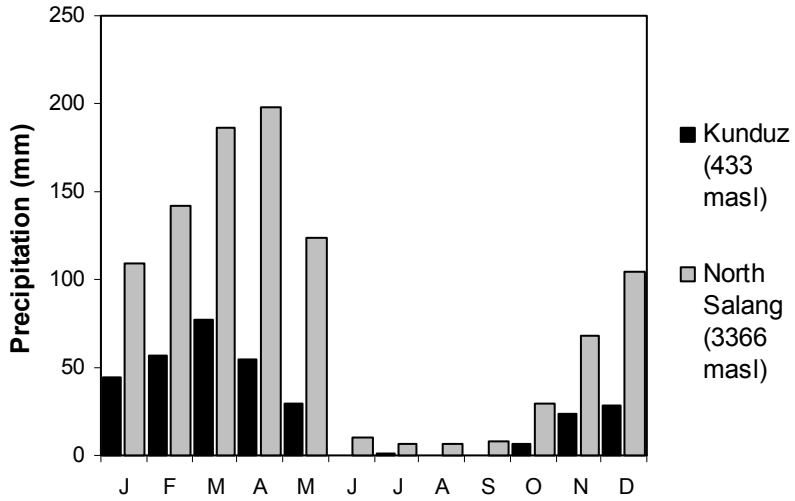


Fig 3: Mean monthly precipitation in the KRB. Variation with altitude: Lowland plains = Kunduz, Upper watershed = North Salang.

The population of the basin is approximately 1.5 million, mainly living on the plains (Beekma and Foxwell 2005). The main economic activity in the basin is intensively irrigated agriculture practised on the highly fertile loess floodplains in the lower reaches of the basin (Riedmann and Beekma 2005). More than half of all irrigated arable land lies north of the main Hindu Kush range in the drainage systems of the Amu Darya River (Alim 2006). The KRB as part of this drainage system is therefore an important 'breadbasket' region for Afghanistan and its management has wider implications for the rest of the country. Indeed, the KRB has often exported crop surpluses to other less productive regions in Afghanistan, although recent problems with security and inter regional infrastructure have reduced its influence in this respect (FAO/WFP 1997).

3.2. Water resources

The basin contains two major rivers: the Baghlan which drains the southerly regions of the basin and the Khanabad which drains the more easterly regions. They merge 10km north of the city of Kunduz and discharge in to the Amu Darya 30km further at the Afghan – Tajik/Uzbek border. The basin follows a rain/snow fed regime (depending on altitude and season) with no glacier-melt input (Aini 2007), although permanent snowfields exist (881 sq. Km).

The rivers within the Kunduz basin maintain a base flow in autumn and winter with peak flow in summer due to the snow pack melting in May and June. The plains receive virtually no rainfall from June to October and are therefore heavily reliant on river flows maintained by snowmelt in the upper watershed areas. In addition annual evapotranspiration rates (1,390mm/yr) far exceed precipitation levels (349mm/yr) per annum (data for Kunduz, altitude: 433masl, Anderson 2006). The hydrological balance is therefore heavily subsidised by upper catchment runoff and groundwater flows. Irrigated fields cover around 2145 sq KM (6.13% of basin) and include 87% of all planted wheat (FAO/WFP 2005). The higher areas in the basin are partly used for rain fed agriculture and mostly consist of deforested lands (Riedmann and Beekma 2005). Table 1 demonstrates the important input of seasonal snowmelt into the basins major rivers during the summer months when precipitation is lowest throughout the basin (fig. 3).

Table 1: Mean monthly discharge (m^3/sec) of the two major rivers in the KRB (see appendix 3 for location).

River	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Baghlan	32	31	31	40	97	235	148	60	39	37	36	34
Khanabad	30	28	32	53	91	186	149	70	36	36	36	35

Cropping patterns are largely governed by the availability of water and in many areas, particularly at higher altitudes only one crop is possible per year. Typically this would be wheat as it is the most important staple food. Other crops include barley, rice and pasture/nitrogen fixing crops such as clover and Lucerne. If there is enough water for a second crop (generally on the plains) vegetables, cotton, mung beans sesame and maize are grown. Cotton and rice are normally only grown in farms with good water supplies as they are water intensive crops (Anderson 2006).

Ground water is found in the flood plains at depths of 5-20m and is significantly fed by infiltration of rivers. There is no tapping of groundwater for irrigation by the traditional system of Karezes (tunnels which transport ground water to the surface) in the KRB and irrigation is mainly done through canal diversion of river flow. Water for domestic use is either taken directly from rivers and canals or accessed via shallow or tube wells (Riedmann and Beekma 2005).

Annual mean runoff for the whole basin is estimated to be 6000M m^3 and irrigation consumption in the KRB is estimated to be about 3,000M m^3. Although this volume is only half of the available runoff, it translates into a constant required flow of 100 m^3/s. However, runoff is distributed extremely unevenly throughout the year as almost 50% occurs over just two months – June and July. It can therefore be said that the KRB experiences water scarcity with demand often exceeding supply (Beekma and Foxwell 2005).

3.3. Water related issues

Environmental Degradation in the Upper Catchment

Deforestation in upper water shed areas has been dramatic over the past 25 years. A UNEP satellite survey of Takhar province in the upper catchment of the Kunduz basin found a decrease from 37% forest cover in 1977 to less than detectable in 2002 (UNEP 2003). The reason for this huge decline in forested areas has been largely attributed to insecure political conditions following the Soviet invasion. Fuel wood was stockpiled as a response to uncertain future access to the forests. In addition military forces reportedly cut trees for their own needs as well as to reduce ambush opportunities for opposing forces (UNEP 2003). Rising poverty and population levels during the decades of conflict produced internally displaced people and refugees whose only means of survival was heavy reliance upon the natural capital around them. Political unrest, poverty and breakdown of community structures tend not to encourage conservation of natural resources. Over harvesting of economically valuable pistachio nuts and overgrazing of livestock in pistachio woodland has reduced forest regeneration. Juniper trees are harvested outright for fuelwood or building materials as there's less economic incentive to conserve intact trees.

Deforestation and overgrazing has reduced the protective vegetative cover of many upper catchment areas leaving soil vulnerable to erosive losses by wind or fluvial action and severely affecting hydrological processes. In addition rising rural populations over the past 2-3 generations has led to increased conversion of former grazing land in upper catchment areas to rain fed wheat crops. Yields are highly variable and crop failure is fairly common. There is extensive environmental degradation resulting from the destruction of original ground cover (UNEP 2003).

Deforested hillsides have accentuated the effects of flash flooding, while riparian deforestation has increased downstream siltation causing damage to irrigation infrastructure.

The Afghan Transitional Authority and the EC agreed that addressing the problems of deforestation and degradation of the upper Kunduz watershed was important to ensure the *"sustainable flow of water to downstream users, as well as addressing the issues of resource access in isolated mountain communities"*. (Beekma and Foxwell 2005)

Groundwater

In the upper catchments rainfed areas clean water for domestic use as well as for agriculture is often a problem as groundwater is too deep to be reached by wells and there are few natural sources or streams. The only sources are flash floods during rainfall and water carried in by donkey. Ice stores (Yachdans) are also utilised which help to preserve winter snows until early summer. These areas are sparsely populated due to low carrying capacity of the land; however increasing numbers of people are forced into these marginal lands due to increasing population pressure (Riedmann and Beekma 2005).

Ground water testing has found faecal bacteria in the ground water of Pul I Khumri – the largest population centre in the Kunduz basin. Here with burgeoning population and a degree of industrial development ground water extraction is increasing while there is no functioning sewer system.

Groundwater tests south of Kunduz city have discovered contamination with traces of herbicides and pesticides from agricultural runoff. This is a good sign that any increased usage as a result of agricultural intensification/extensification could have severe implications for the purity of groundwater supplies for human consumption (Riedmann and Beekma 2005). Other groundwater issues include a high degree of salinisation both in Kunduz and Takhar province due to wash out of salts from base rocks. Raised water tables through irrigation, and high evaporation rates work to bring salts from limestone base rocks to accumulate in the upper soil profile. Agricultural inputs and fertilisers can add to this problem. In addition a lack of rainfall and limited drainage means there is little leaching through the soil profile and hence salts collect in the upper profile (Anderson 2006).

Flooding/drought

All though flooding and drought are opposing phenomena in impact, they both have similar root causes and both cause a great deal of damage to livelihoods in the Kunduz basin. Seasonality and unpredictability of precipitation is the major driver in both cases. As in 1999-2001 low winter precipitation combined with above average temperatures resulted in an insufficient snow pack, earlier runoff and drought conditions in the summer months. Low spring rainfall intensified the problem. On the other hand coincidence of heavy rains with spring melt of the snow pack leads to flood conditions which are often extremely damaging to irrigation infrastructure and agricultural land. This range of river discharge magnitudes which irrigation systems must cope with poses a real challenge to farmers in the KRB (Anderson 2006). Both drought and flood events are exacerbated by deforestation in upper watershed regions. Forest cover often helps to increase infiltration which reduces flood causing surface runoff as well as forming a slow release mechanism to recharge groundwater stores and thus counteracting drought conditions (Dijk and Keenan 2007). However, forest-hydrology interactions are complex and therefore site specific studies are required to reach robust conclusions.

Water management structures

Efficiency of water use and distribution within the basin is a major issue (Aini 2007). Overall efficiency of traditional surface irrigation is poor at around 30% (Qureshi 2002). These may be to a large degree social issues such as lack of coordination during cultivation and when and how water is distributed. Afghanistan has an ancient, well regulated water management tradition and indeed social water management networks exist in the KRB, as elsewhere in Afghanistan. These are headed by a community elected Mirab (watermaster), who overseas operation and maintenance of irrigation systems. However, the past few decades of conflict have served to disrupt traditional social structures and community cohesion that ensured well distributed efficient use of water resources. Conflict, abuses of warlords, emigration, and drought induced land abandonment have all contributed to this process. While equitable distribution of resources depends upon the reinstatement of community management structure, efficient use of resources can be enhanced by rehabilitation of irrigation structures (desilting canals etc.) as well as possibilities of water harvesting at peak flows (Qureshi 2002).

3.4. Institutional analysis

Defeat of the Taliban in 2001 led to a complete restructuring of government. New ministries were created tasked with the environmental challenges in Afghanistan, namely the Ministry of Agriculture and Animal Husbandry, Ministry of Irrigation, Water Resources, and Environment, Ministry of Rural Rehabilitation and Development which all have remits in the sphere of water management albeit with limited resources (Qureshi 2002).

Several NGO's operate in KRB such as Mercy Corps, German Agro Action and the Aga Khan Foundation as well as other multilateral donors such as the EU and the World Bank. The German military are in charge of security in Kunduz and contribute to development projects through its Civil Military Cooperation Programme, as well as providing the security necessary for other work to take place (Beekma and Foxwell 2005).

Integrated Water Management at the river basin level has been chosen as the basis for the development of water resources policy in Afghanistan. The Kunduz River Basin Programme is the first to undertake a river basin management approach as it is still a fairly new concept in Afghanistan (Beekma and Foxwell 2005). The basis of this approach is that the river basin is recognised as an interconnected hydrological system with corresponding cause and effect relationships across the natural drainage area both temporally and spatially. This approach is integrated into a framework of improved institutional structures and good governance practice and social development.

While policy formulation and development efforts are currently domestically orientated in Afghanistan, an important future dimension is the KRBs discharge to the Amu Darya River and by extension the Aral Sea

crisis (Micklin 2007). Other Central Asian countries formed an agreement in 1993 to address the issue but little serious progress has been made. Planned intensification of irrigated agriculture in the Northern plains of Afghanistan is essential for the food security of Afghanistan but will only contribute to the continuing crisis of the Aral Sea. Afghanistan has started to draw up its transboundary water policy as of April 2007 in which it hopes to create water sharing treaties with its neighbours over future projects such as dams which will affect downstream supply (Fipps 2007). Future international treaties may therefore play an important role in the future of water policy in the Amu Darya drainage basin of Afghanistan (Nakayama 2003).

4. DATA AND METHODOLOGY

The overall approach to this project has been that of a basin level study. As such most data has been interpreted or converted to the basin level. This is largely due to the resolution of historical snow cover data being too coarse to give accurate enough results at the sub–basin level. It is therefore important to remember that there exists a good deal of heterogeneity within the basin which will inevitably be masked by this approach. However it does give a synoptic view of the basin as a whole which provides a baseline for further study.

4.1. Climatology Data

The CRU TS 2.1(Mitchell and Jones 2005) data-set comprises an interpolated 1224 monthly grids of observed climate, for the period 1902-2002, and covering the global land surface at 0.5 degree resolution. This was used to obtain both a mean monthly precipitation and temperature time series for the upper watershed regions of the Kunduz basin (appendix 1).

The mean values of a three grid sample were used to provide a single winter (DJF - December to February) and spring (MAM – March to May) average value for precipitation and temperature from 1902-2002. The mean elevation of grids was 3411masl. A single value was necessary as the basin was the unit of analysis employed for the snow cover dataset. This climatology was used to examine the main snow accumulation season, (December - February) and ablation period (March-May).

4.2. Satellite Data

Two satellite datasets were used in this study to provide a single snow cover time series spanning the period 1972-2007 (appendix 2).

1) The Northern Hemisphere EASE-Grid Weekly Snow Cover

The EASE-Grid Weekly Snow cover from the National Snow and Ice Data Center *(*Armstrong and Brodzik. 2005) provided snow cover data from 3 October 1966 through to 5 June 2002 in weekly intervals. These are digitized versions of NOAA/NESDIS Northern Hemisphere Snow Extent charts. Data are provided in a binary format, indicating snow or no snow in the Northern Hemisphere 25 km equal-area grid (NSIDC EASE-Grid) and obtained by the AVHRR sensor since 1972. Snow cover presence or absence within a grid cell is determined on the basis of the last observed clear day within the observation week, this minimises cloud interference in the dataset. Furthermore, snow cover in areas obscured by cloud in the resulting map is estimated using passive microwave data (SSM/I); therefore the available product is cloud free. Data from December to May was analysed to cover the main winter season and spring ablation period.

2) MODIS/Terra Snow Cover 8-Day 500 m Grid

Snow cover products from the Moderate Resolution Imaging Spectro-radiometer (MODIS) provide data at 8 day intervals at 500m resolution for snow seasons (September to May) 2002 - 2007. The data represents the maximum snow cover extent for the 8-day composite period, meaning that if a pixel was classified as snow in any of the 8 days; it will be classified as snow in the composite product. This again minimises cloud cover problems. However cloud isn't completely screened out of the MODIS product so any results with cloud cover >5% were omitted from the dataset. Only data for December - May were used as above.

Constructing the Snow Cover Time Series

Modis and EASE grid snow data were analysed and recorded as percentage cover of the basin in all regions above 2500masl and then compiled to form a time series from 1972-2007. The year value refers to January – May section of snow season but includes December of the previous calendar year. The Ease grid provided data 1972-2001, Modis dataset was used for 2002-2007. Data collection improved markedly due to the introduction of the Advanced Very High Resolution Radiometer (AVHRR) sensor in 1972, pre-1972 data are therefore omitted due to inherent inconsistencies between data (after Bamzai 2003).

Defining Region of Interest

A Vector layer was used to subset the Kunduz River Basin (Source: Mike Budde, USGS FEWS-NET)

Elevation dataset SRTM (Shuttle Radar Topography Mission) GTOPO30 (Global Land Cover Facility http://glcf.umiacs.umd.edu/index.shtml) was applied to interrogate regions above 2500masl, which represents 15400 square Kilometres (44% of basin). This was important as snow cover is much more transient below this elevation in the Kunduz River Basin and would introduce too much variability. In addition snow cover below this elevation contributes to a lesser degree to the snow pack which feeds spring/summer river flows. All snow cover results have been interpreted as percentage cover of all areas greater than 2500masl in the Kunduz Basin.

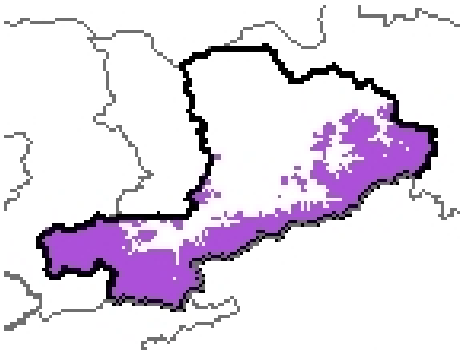


Fig 4: layer of regions > 2500masl used to interrogate snow cover data

Snow Water Equivalent and Snow Depletion Curves

Snow water Equivalent (SWE) is expressed as the standing column of water resulting from the melting of a snow sample equal to the depth of the snow pack. This is one of the most important factors in determining available water released by a melting snow pack. Unfortunately there are few data from direct measurements in the Kunduz basin.

A commonly used method used to approximate SWE in the absence of field data involves constructing snow depletion curves which plot snow cover area (SCA) against time. This method is based on the fact that a thin snow pack (low SWE) starts depleting early and also depleting faster compared to a thick snow pack (high SWE) which starts depleting later and also depletes slower. In general, the depletion curve is the cumulative effect of the SWE in the basin, the temperature regime and the under lying terrain (Shamir and Georgakakos 2007).

I have used snow depletion curves in this project in a qualitative manner in order to add robustness and support other results.

4.3 Hydrology Data

Mean monthly river discharge data was obtained from 14 gauging stations across the Kunduz River Basin covering the Baghlan, Taloquan and Doshi Sub-basin. The data was collected between 1965 and 1980 (appendix 3).

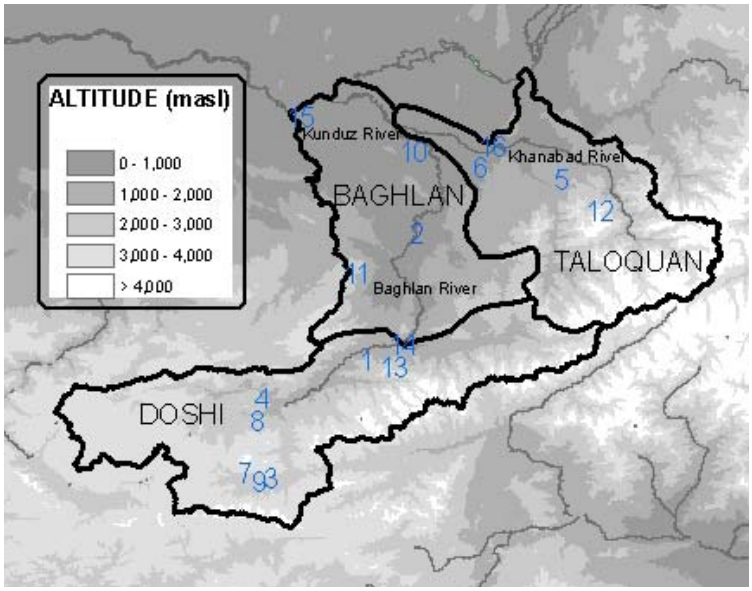


Fig 5: Position of gauging stations within the Kunduz River Basin. All three sub-basins (Baghlan, Taloquan and Doshi) are well represented.

Mean June discharge data was selected for analysis as peak discharge occurred in this month (13 of 14 records) following the main snow ablation season of March – May. These data were converted into a discharge anomaly figure by calculating the percentage deviation from mean June discharge of the 16 year record. This made it possible to analyse, as a group, river discharge rates which displayed a great deal of variation in magnitude (range 1.32-324.7m^3/s mean 128.2 m^3/s).

A composite Basin Discharge Anomaly (BDA) was obtained from the mean of all individual station discharge anomalies. This was done because the unit of analysis in terms of snow cover was at the basin level. Therefore it was important to produce a composite figure which reflected the hydrology of the basin as a unit. The number of stations contributing to the BDA varied in any given year (see Appendix III, Mean number of stations = 10)

4.4 Vegetation Indices

Crop Production Data

Crop production data is available for the North East region of Afghanistan from joint FAO/WFP reports. Irrigated wheat data was used to provide an index of climate/agriculture interactions as its production is heavily dependant on snow melt fed canals and additionally wheat is the most important agricultural product in Afghanistan.

It is important to note that the wheat figures represent the whole of the North East of Afghanistan which includes the province of Badahkshan to the east of KRB; however Badahkshan only represents about 10% of the irrigated wheat figure so will have a limited leverage effect on the data.

Landsat/NDVI

The Normalized Difference Vegetation Index (NDVI) is used as an index of vegetation density/health. NDVI is calculated as a difference between the reflectances of red and infrared wavelengths. During the process of photosynthesis in leaves red wavelengths (Landsat band 3) are absorbed and infrared (Landsat band 4) are reflected, so the difference between the two is correlated with the amount of photosynthesis and therefore a measure of primary production.

$$NDVI = IR\text{-}R \,/\, IR\text{+}R$$

Where:

IR = Infra red (band 4)

R = Red (band 3)

NDVI values range from -1 - +1, higher positive values signify more dense/healthy vegetation; negative values generally represent absence of vegetation.

Landsat scene (path 153 row 35 WRS-2) was used to calculate NDVI values in 1990 and 2001. This allowed a comparison of the drought year 2001 with control year of 1990. Its purpose is illustrative rather than analytical.

4.5. Climatic Change Projections

Country level temperature and precipitation data were obtained for Afghanistan from the Climate Research Unit, UK (TYN CY 2.0, Mitchell et al. 2002). The data used were from the HadCM3 Global Circulation Model run by the Hadley Centre, UK. The values represent the changes between 1961-90 and 2070-99 and are available for 2 seasons, winter (December – February) and summer (June - August) and 2 SRES emission scenarios (A2 and B2).

The SRES Emissions Scenarios:

A2 – "A very heterogeneous world. The underlying theme is that of strengthening regional cultural identities, with an emphasis on family values and local traditions, high population growth, and less concern for rapid economic development."

B2 – "A world in which the emphasis is on local solutions to economic, social, and environmental sustainability. It is again a heterogeneous world with less rapid, and more diverse technological change but a strong emphasis on community initiative and social innovation to find local, rather than global solutions." (Nakicenovic and Swart 2000))

4.6. Statistical Analysis

1) The correlation test used throughout this report is Pearson's product moment coefficient. It seems that correlation results down to r = 0.5 are considered reasonably strong in the snow cover literature (Qian and Saunders 2003, IPCC 2007, Dahe 2006),

2) Simple linear regression was used to generate an equation which describes the line of best fit that most accurately reflects the relationship between two variables. This was used to extrapolate observed trends into the future in order to generate some idea of how climate change scenarios will impact the kunduz Basin. Much caution is necessary when interpreting these results as relationship between just two climatic parameters tend to be relatively weak due to the complexity of interactions that are not included in a simple linear relationship. Therefore errors or inaccuracies would be amplified as a detected trend is projected into the future. With this important point in mind there is some use in this exercise in order to compare the relative impact of climatic parameters within an order of magnitude.

3) Un-paired T –test was used in order to test for difference between the means of two decades of snow data 1972-81 and 1998 – 07.

5. RESULTS AND ANALYSIS

5.1. Analysis of Climatic Trends

CRU temperature dataset

A positive linear trend was found in decadal mean temperatures of the KRB in both winter (DJF fig. 6) and spring (MAM fig. 7) over the period 1902-2002. Linear trends for DJF and MAM show an increase of 1.37ºC and 1.32ºC respectively over the 1902-2002 period. MAM temperature increase was significant at the 99% whereas DJF temperature increase was only significant at the 90% level. Additionally, the correlation of MAM decades was stronger (r=0.83) than DJF decades (r=0.62). There was a greater degree of inter-decadal variation in winter temperatures (range 2.29ºC) than spring temperatures (range 1.74). Overall increased temperature in MAM is a much more robust trend.

90% of winters and 80% of springs were anomalously warm over the last decade of analysis (1993-2002). Full time-series for temperature and precipitation are given in appendix 2.

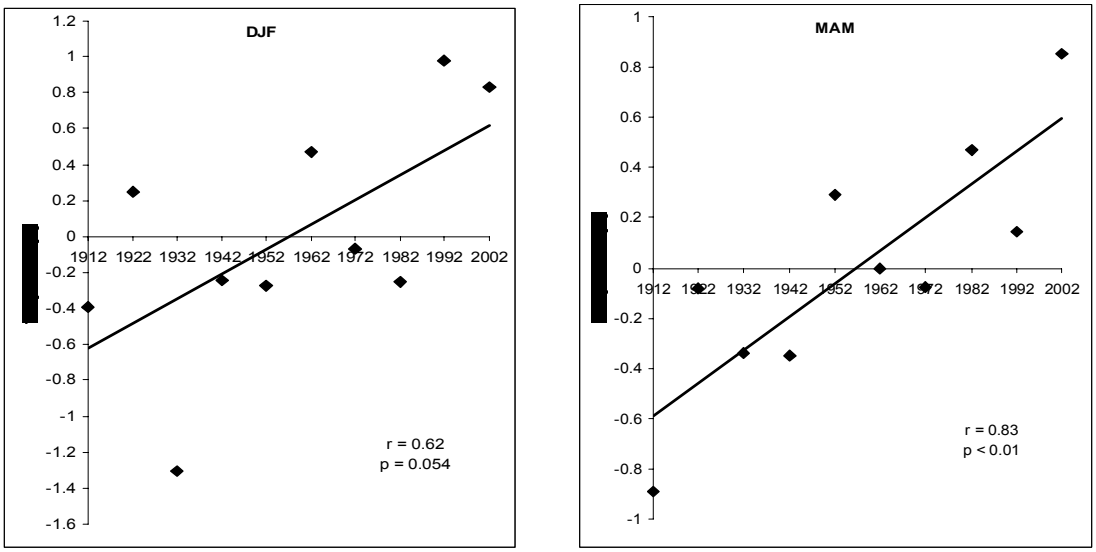


Fig. 6 and 7: Decadal mean temperature anomalies for winter (DJF) and spring (MAM). Date refers to end date of decade for which a mean is presented. DJF increase significant at 90%, MAM increase significant at 99% level. Linear trend shows greater absolute increase DJF (1.37ºC) than MAM (1.32º).

CRU Precipitation Dataset

In the analysis of CRU precipitation data (fig. 8) no significant trends where found for winter (DJF), Spring (MAM) or Summer (JJA). A significant increase in Autumn (SON) rainfall was found ($r = 0.80$, $p<0.01$) with a positive linear trend of 57% over the period 1902-2002. Further analysis suggests a positive linear trend in Spring (MAM) of 22% as all 5 decades since 1952 were found to have above average precipitation (1902-2002 mean = 74.3MM) and 4 of 5 pre-1952 decades were found to have below average precipitation.

However a high degree of variation around the linear trend line (r=0.45) contributed to making this a non-significant result. However, due to the magnitude of spring precipitation it is potentially an important result. Close inspection of figure 6 suggests a multi-decadal oscillation in precipitation patterns probably as result of its close association with the cyclical natural phenomena of ENSO/NAO (Syed 2005).

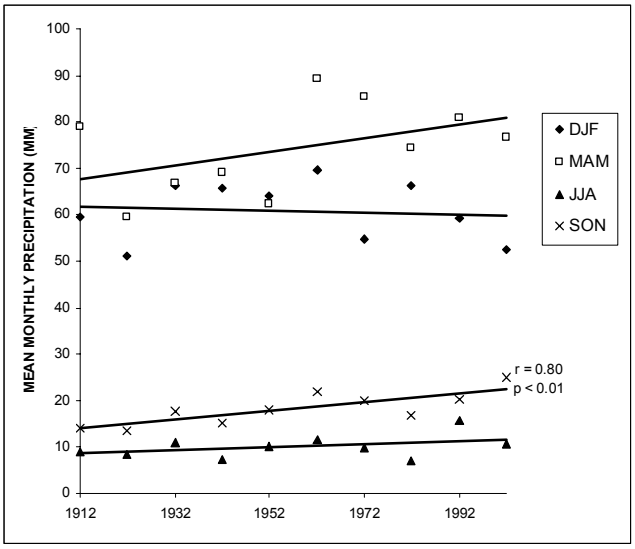


Fig 8: Only SON significant. Suggests DJF not increasing as stated in FAR WGI 62% increase SON 1902-2002.

Snow Cover Area

Snow cover above 2500masl in the Kunduz River basin was analysed by satellite data over the period 1972-2007 between December and May each year (see appendix 2 for time series).

a) Snow cover and temperature

A significant negative (weak) correlation was found to exist between mean spring snow cover (MAM) and mean spring temperature over the period 1972-2002 (fig 9: $p < 0.05$). However there was a good deal of variation around the trend line ($r = 0.55$) as you would expect because temperature is not the only variable determining snow cover extent (i.e. precipitation). No significant correlation was found between mean winter snow cover and mean winter temperatures over the same period. This is probably due to the greater sensitivity of spring snow cover to temperature increases compared to winter cover as spring snow is generally closer to its melting point with temperatures higher in spring than winter.

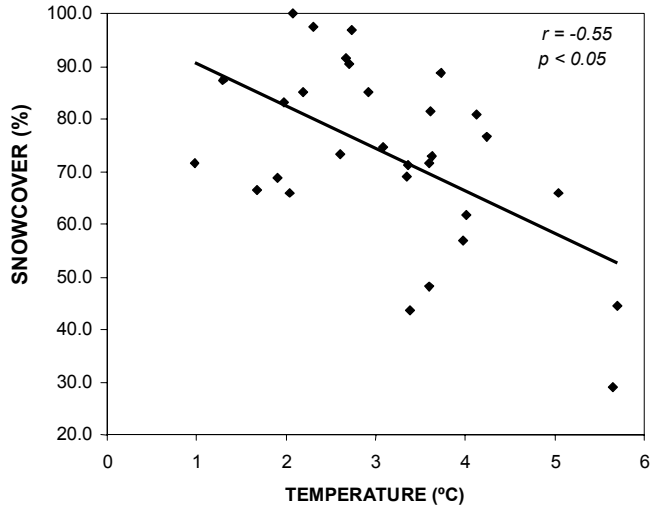


Fig 9: Correlation between mean spring (MAM) snow cover and mean temperature over the period 1972-2002. A negative correlation was found to be significant at the 95% level with snow cover decreasing with elevated temperature.

b) Snow cover and precipitation

A highly significant positive correlation was found between MAM snow cover extent and mean MAM monthly precipitation (fig 10: $p < 0.001$). Again the strength of the correlation ($r = 0.61$) is probably reduced by other factors involved in determining snow cover (i.e. temperature).

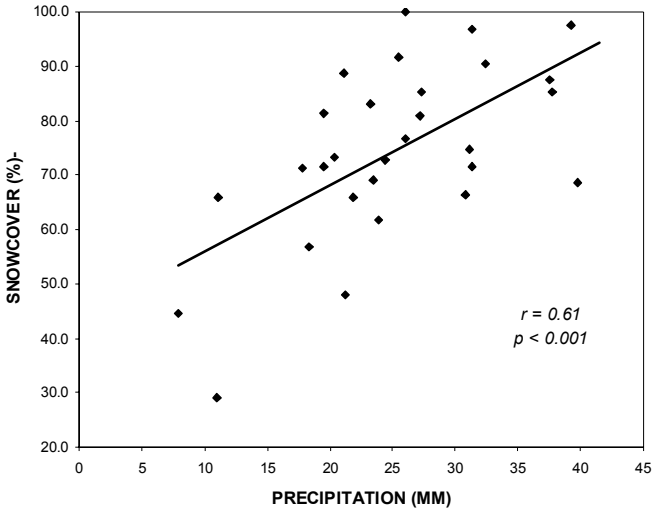


Fig 10**:** Correlation between mean MAM snow cover and mean MAM precipitation over the period 1972-2002. A positive correlation was found to be significant at the 99% level with snow cover increasing with increasing precipitation.

5.2. Analysis of Current Snow Cover

Inter-Decadal Change in Snow Cover Area

Snow cover has decreased by 10% on average in the KRB over the whole snow season (December – May) between 1972 and 2007 (fig 11). This constituted a 4% reduction in winter snow cover and a much greater 15% reduction in spring. This is due to the greater correlation between spring snow cover and temperature and non-correlation between winter snow cover and temperature. An unpaired t-test revealed the difference between mean 1972-81 snowcover and mean 1998-2007 spring snow cover to be significant at the 95% level. The decrease in winter cover wasn't found to be significant.

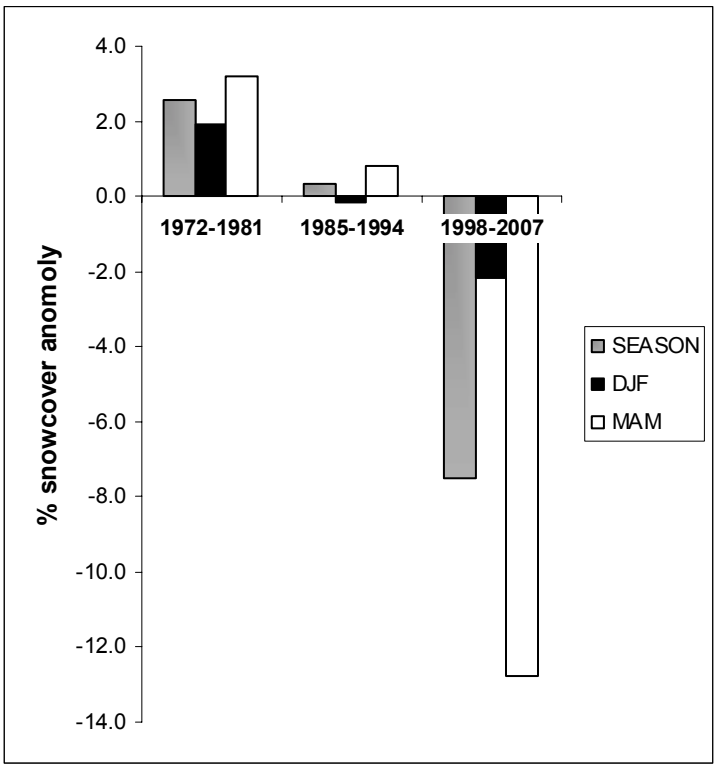


Fig 11: Decadal % snow cover anomaly from long term mean (1972-2007) for Dec-Feb (DJF), Mar - May (MAM) and whole snow season Dec-May.

Intra-Annual Variation in Snow Cover

The percentage snow cover was found to deviate from the long term mean to an increasing extent as the snow season progressed (fig. 12). Critically, mean monthly snow cover extent of the last ten years shows that snow cover has been below the long term mean in every month. The minimum anomaly was in January (-1.6%) and maximum in May (-33.5%).

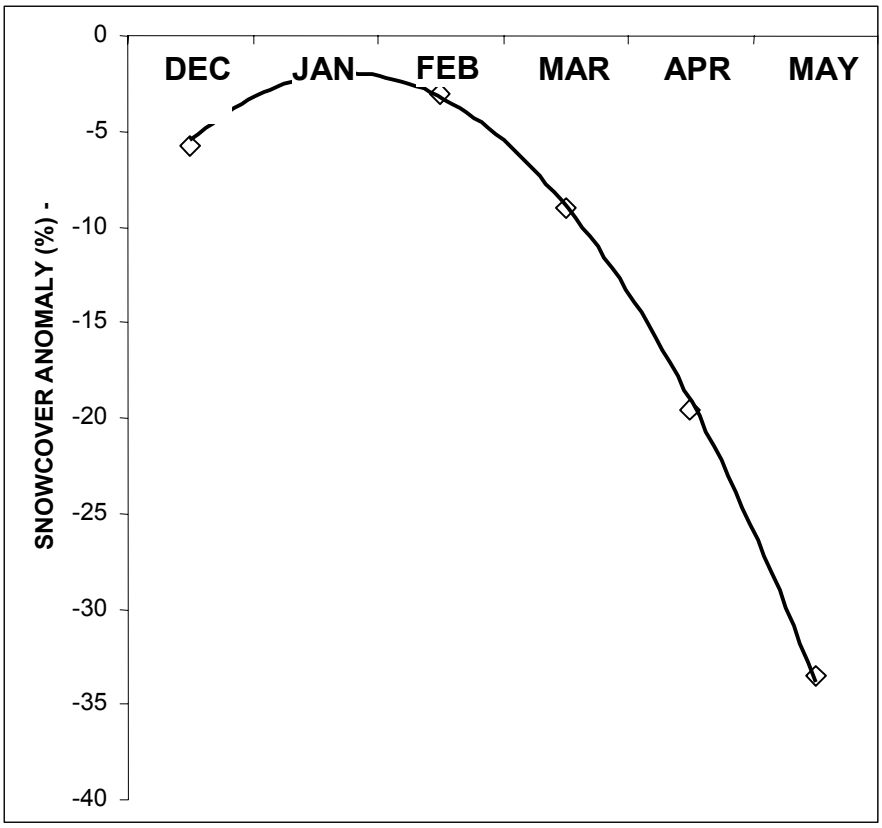


Fig 12: Mean deviation of monthly snow cover extent from long term mean (1972-07) for the last decade (1998-07). All months are below mean snow cover extent. The snow cover anomaly increases from a minimum in January (-1.6%) to a maximum in May (-33.5%).

5.3. Impacts of Reduced Snow Cover Extent

Snow Cover and Basin Hydrology

The most significant correlation was found to exist between mean March – May (MAM) snowcover (% >2500masl) and the basin discharge anomaly (BDA – see methodology for discussion) (fig. 13: r = 0.81, p< 0.01). There was no correlation found between individual winter months snow cover or mean December to February cover (DJF). A correlation was found between both March and May snow cover (r = 0.78 and 0.72 respectively) and this was significant at the 95% level. Surprisingly perhaps, very little correlation was found between April snow cover and BDA (r = 0.32). Mean March – May (MAM) snow cover is therefore considered in this analysis to be the most useful indicator of actual available water.

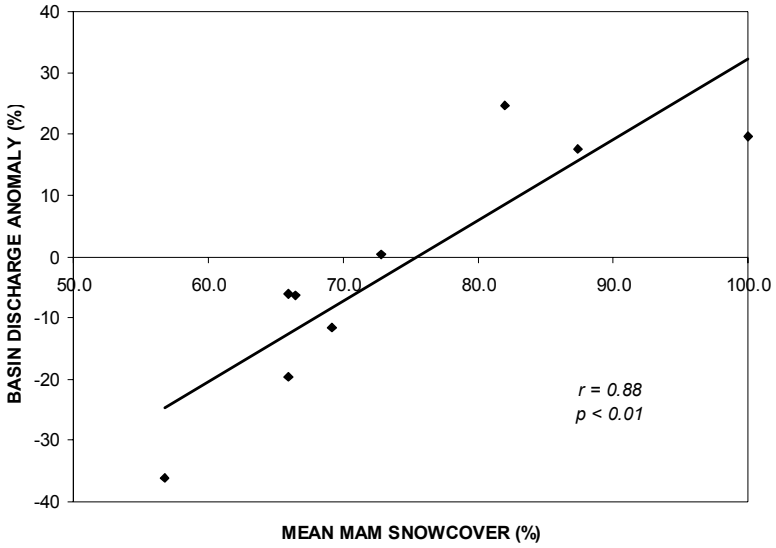


Fig 13: Correlation between mean March – May (MAM) snow cover area and basin discharge anomaly (BDA), significant at the 99% level.

Conducting a linear regression gives the expression;

y = 1.3123x - 99.126

- where y = basin discharge anomaly (BDA %) and x = mean snow cover March – May (%). Using this empirical relationship BDA can be expressed as a function of snow cover (Fig 14). This enables snow cover time series to be analysed more usefully in terms of the more relevant basin discharge figure.

Normal basin discharge would be expected to be associated with normal snow cover levels. Therefore this relationship is verified to some extent by the fact that the discharge anomaly is 0 at 75% MAM snow cover according to the linear expression; and actual mean MAM cover is relatively close at 70%.

An analysis of the literature reveals a good association between this simulated BDA (fig. 14) and actual drought reports from the KRB. The severe drought of 1999-2001 (FAO/WFP 1999, 2000, 2001) is well represented with BDA falling sharply between 1999 and 2000 to a low in 2001. The BDA in 2001 was minus 55% which helps to explain the severity of the drought. 2003 was a strong recovery year (FAO/WFP 2003), in hydrological terms with good precipitation and represents a peak (albeit still below the mean). 2004 was again a slump dry year (FAO/WFP 2004) followed by strong recovery in 2005 (FAO/WFP 2005) but again slumps in both 2006 (MRRD/WFP 2006) and 2007 (MAIL 2007). A drought in 1977-78 (IWMI 2002) is illustrated by a BDA of -10% although this figure isn't particularly low, below average BDA since 1974 possibly had an accumulative effect in reducing soil moisture. The last most significant drought before this recent one was in 1970-71 (IWMI 2002) and is obvious with a BDA down to approximately -40%. The 1986-87 spike remains unexplained and possibly represents a problem with the snow cover dataset as no drought was reported by the literature at this time. Most relevant is the fact that these figures show consistently below average simulated BDA over the period 2000-2007 and indicates the most sustained period of water stress for at least 38 years.

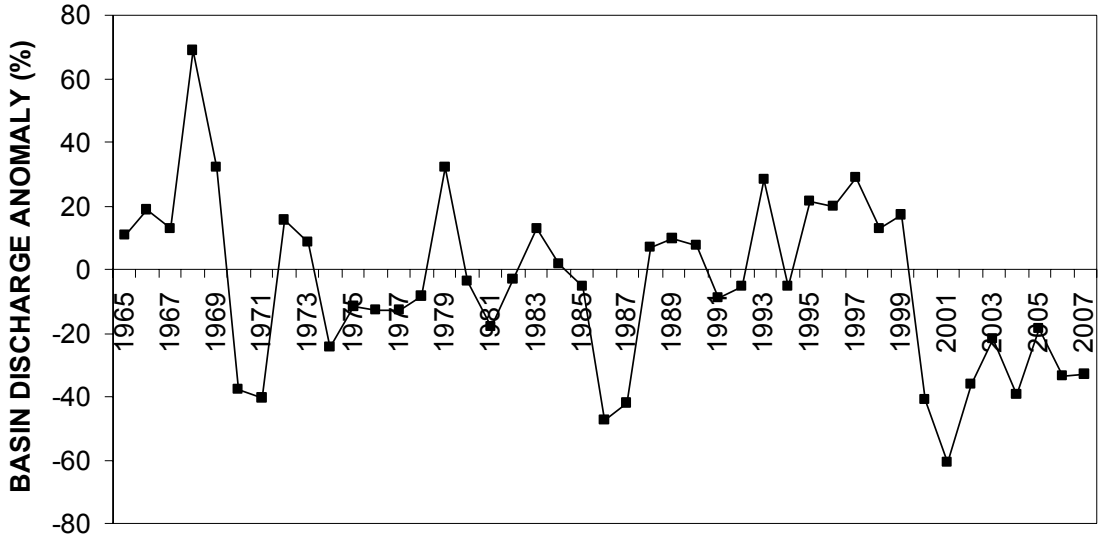


Fig 14: Composite Basin Discharge Anomaly (BDA) time-series for the Kunduz River Basin as a single hydrological unit. 1965-1980 field observations. 1981-2007 based on regression analsysis. Droughts in 1970-71, 1977-78 and 1999-01 are all associated with negative BDA.

Snow cover and Agriculture

A highly significant correlation was found between snow cover extent and irrigated wheat area (FAO/WFP 2000-05, MRRD/WFP 2006, MAIL 2007) (fig 15: $r = 0.86$, $p < 0.01$). It is important to note that the wheat figures represent the whole of the North East of Afghanistan which includes the province of Badahkshan to

the east of KRB; however Badahkshan only represents about 10% of the irrigated wheat figure so will have a limited leverage effect on the data. Data availability was limited to 8 years.

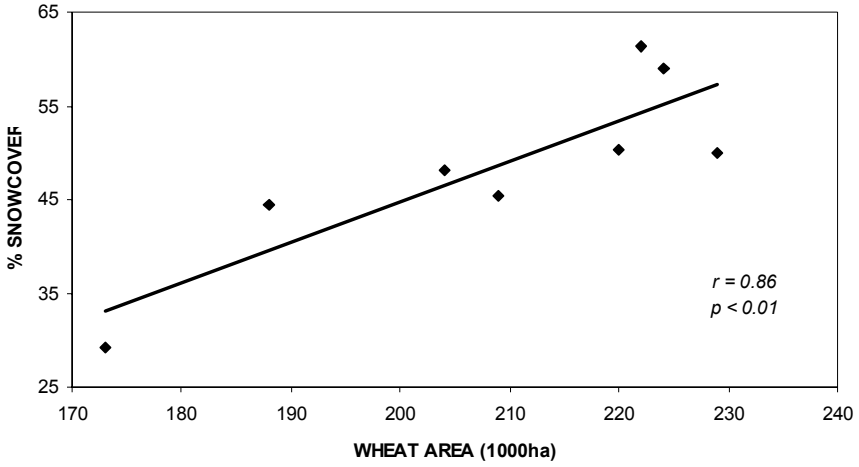


Fig 15: Correlation of % March-May mean snow cover with irrigated wheat area planted 2000-2007($r = 0.86$, $p<0.01$ $n = 8$).

- Snow depletion curves

Snow depletion curves indicate depth of snow pack (and hence snow water equivalent) by the rate of snow cover decline (steepness of slope) and magnitude of start point. This is based on the premises that a thin snow pack (low snow water equivalent) starts depleting early and also depleting faster compared to a thick snow pack (high snow water equivalent) which starts depleting later and also depletes slower .

An analysis of the curves in figure 16 in conjunction with irrigated wheat production data helps to corroborate the correlation described above (fig 15). Bad wheat years of 2001 and 2003 (258 000 and 466 000 tonnes respectively) are clearly distinguishable from good wheat years of 2003 and 2004 (731 000 and 561 000 tonnes respectively) by much reduced area under the depletion curve.

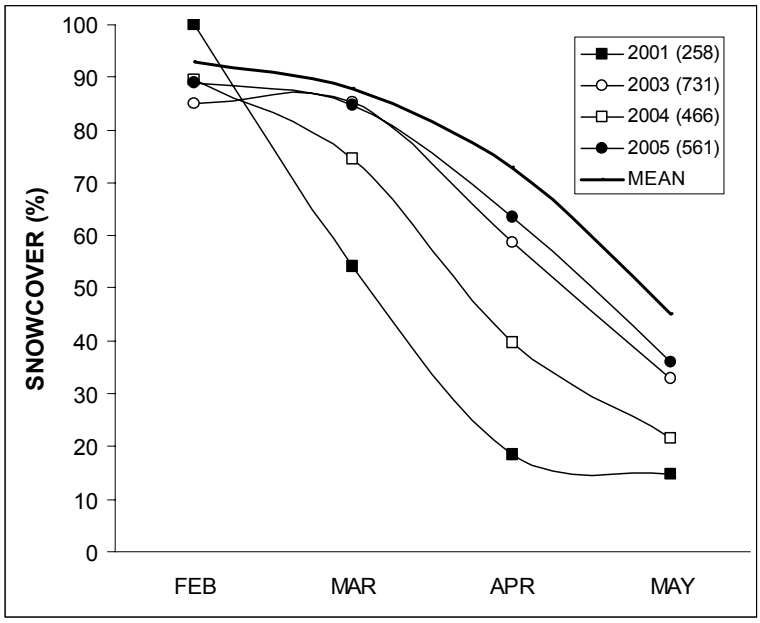


Fig 16: Snow depletion curves of 2 good irrigated wheat production years (2003, 2005) and 2 bad wheat years (2001, 2004). Wheat production figures given in brackets (1000 tonnes).

Case study: Drought 1999-2001

Anomalously high temperatures and low precipitation led to widespread drought from 1999 – 2001 (fig 17). In spring mean temperatures in 2000 and 2001 were the highest in 100 yr record (appendix 1) and this contributed to early melt of a shallow snow pack (fig 16) which had suffered from low winter precipitation. Spring mean snow cover fell to lowest extent in 35 year record (29.1%).

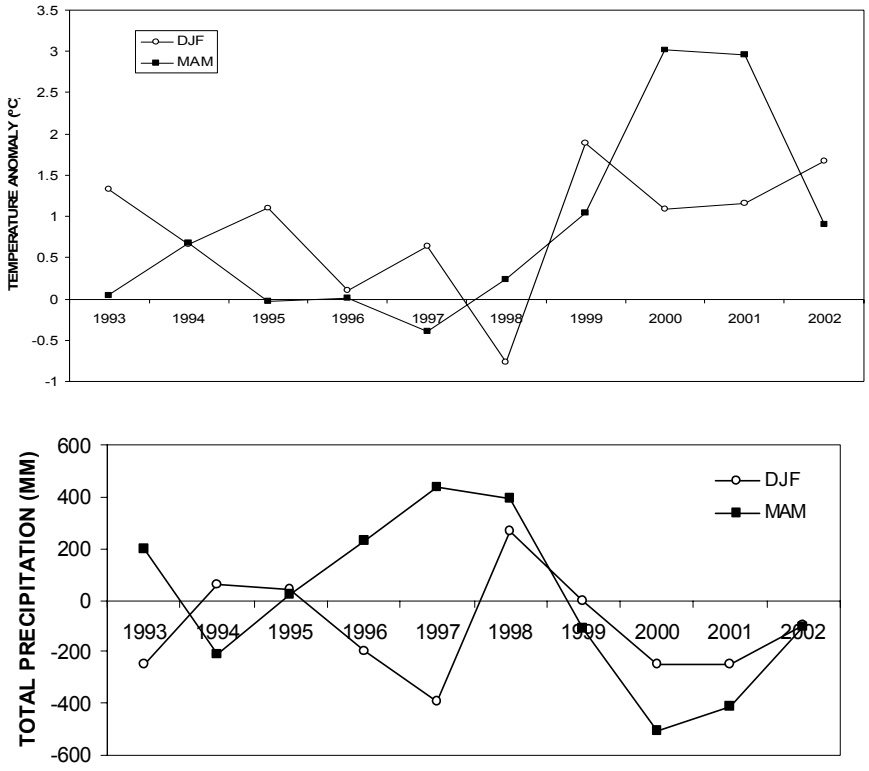
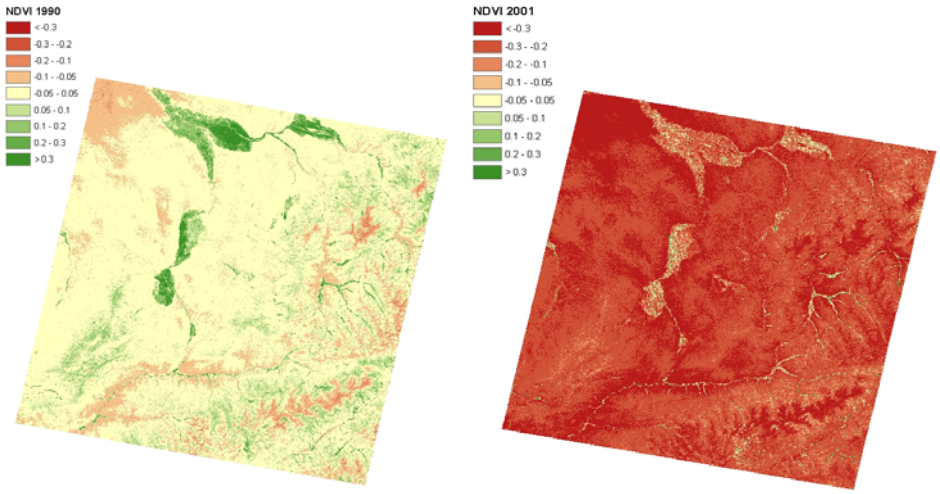


Fig 17: Ten year record of winter and spring temperature and precipitation anomalies. High temperature combined with low precipitation led to widespread drought 1999-2001. Temperature of 2000 and 2001 highest of 100 year record. 90% DJF temperatures > mean, 80% MAM temperatures > mean over period 1993-02.

The effect of this drought on vegetation in the Kunduz basin is well illustrated by an NDVI analysis of landsat images from summer of 1990 (control MAM snow cover anomaly = +12.1%) and 2001 (MAM

snow cover anomaly = -40.2%). Positive NDVI values represent vegetation; higher values signify healthier more dense vegetation. Negative values represent bare ground (fig 18).

While this drought was a regional phenomenon and driven by la Niña and associated cyclical climatic forcing, anomalously high temperatures undoubtedly exacerbated the situation and reduced snow cover to an even greater extent than low precipitation levels alone would have done, resulting in widespread summer water stress.



NDVI: 3/8/1990

>0.0 = 20.8%

>0.1 = 4.45%

>0.2 = 2.02%

>0.3 = 0.82%

>0.4 = 0.08%

NDVI: 8/7/2001

> 0.0 = 0.80%

>0.1 = 0.38%

>0.2 = 0.17%

>0.3 = 0.06%

>0.4 = 0.01%

Fig. 18: NDVI results for control year of 1990 and drought year of 2001. Percentage of Landsat scene with NDVI greater than given value. Impact of the drought on vegetation is clearly seen 1990 4.45% NDVI > 0.1 / 2001 0.38% NDVI > 0.1.

5.4. Future Climate Projections

A regression analysis of both temperature and precipitation in respect to snow cover from 1972-02 gives equations for the lines of best fit *(see fig 4 &5).*

1) Temperature; $y = -8.0441 \chi + 98.568$

2) Precipitation; $y = 1.2258 \chi + 43.587$

- where y = snow cover; χ = temperature/ precipitation.

An approximation of the effect of future climate change on snow cover can then be made based on IPCC emission scenarios A2 and B2. The data inputted was from TYN CY 2.0 dataset obtained from the Climate Research Unit, UK (Mitchell *et al.* 2002) and based on the Hadley Centre model (HadCM3). Although relatively crude, this modelling process provides some idea of the relative importance of projected changes in temperature and precipitation on water availability based on observed relationships over the period 1972-02.

Table 2: Relative impact of future temperature and precipitation regimes on snowcover in the KRB > 2500masl

IPCC emission scenarios	TEMPERATURE (ºC)	% SNOW COVER CHANGE	PRECIPITATION (MM/%)	% SNOWCOVER CHANGE	%TOTAL CHANGE
A2 SRES	+4.50	-36.2	-1.30%	-0.4	**-36.2**
B2 SRES	+3.55	-28.6	+14.08%	+4.3	**-24.22**

An evaluation of this linear relationship:

Mean MAM temperature 1972-2002 is 3.12ºC. Model = 73.5% SCE

Mean MAM precipitation 1972-2002 is 25.6mmn. Model = 74.23% SCE

Actual SCE= 69.8 %

Approximately 4% overestimate of SCE by model

It appears that likely future impacts of changing precipitation will have a small negative or positive impact on snow cover giving a likely range of -0.4% to +4.3%. Projected temperature changes are likely to have a much greater impact with a likely range of a -28.6 to -36.2 % reduction in snow cover.

Furthermore, if a starting point of 56.1% SCA (1998-07 mean) is taken; 19.9% - 31.88% is the projected range of likely snow cover area by the end of 21st century. This figure should be compared with the fact that during the 2001 drought year the mean spring snow cover was down to 29.1% with serious consequences. Although this analysis is rather simplistic it does give a rough idea of the relative magnitude of impact likely changes in temperature and precipitation will have on snow cover.

5.5. General Limitations

Many of these results are based on correlations between two fairly variable environmental factors (i.e. snow cover and temperature). The correlations tend to be much weaker than those you would expect from more controlled experiments such as those in biological sciences due to the natural variation that exists year to year as well as the complex interaction of climatic parameters at the basin level. Similarly, regression analyses based on weak correlations (whatever the p – value) are naturally prone to increasing error as you extrapolate a trend into the future. However, they do give an indication within an order of magnitude about the relative influence of different environmental factors in a future climate scenario.

Basin as unit of analysis

This is a basin wide analysis and does not account for climatic heterogeneity between sub basins. However this is unlikely to be of a significant concern in the long term as average precipitation and discharge figures for the two upper catchment sub basins, Taloquan and Doshi, are fairly similar.

Table 3: Mean precipitation of two upper catchment sub basins and discharge of their main rivers.

Sub basin	Mean precipitation (Km3)	Main river	Discharge (m3/sec)
Doshi	3.43	Baghlan	68
Taloquan	3.76	Khanabad	65

However, where this difference is key is on year to year monitoring where a good deal of heterogeneity is evident. For example Table 4 shows snow water equivalent values for the two sub basins; 2005-06 Taloquan has double SWE of Doshi whereas 2006-07 they are relatively equal.

Table 4: Interannual variability of snow water equivalent (SWE) in two upper catchment basins.

Sub basin	Mean precipitation (km3)	SWE (Km3) 2005-06	SWE (km3) 2006-07
Doshi	3.43	0.48	0.608
Taloqan	3.76	1.06	0.647

Data limitations

The snow cover time series is relatively short and attempts to quantify change in environmental systems that have been shown to operate at scales of decades or so. For example, both temperature and precipitation have demonstrated a great deal of variability at the decadal scale (examples. In addition the causal nature of this relationship needs further testing as there appear to be a feedback mechanisms between snow cover and surface air temperature with greater snow cover causing an enhanced albedo effect which serves to reduce surface air temperatures. Therefore, conversely, a reduction of snow cover would contribute to increasing near surface temperatures. Therefore the cause and effect nature of this relationship needs to be established.

6. DISCUSSION

6.1. Observed Climatic Trends

It appears to be highly likely that the upper catchment areas of the Kunduz River Basin experienced a rise in temperature of 1.32ºC over the 20th century during the spring months of March – May. There are indications that a temperature rise of 1.37ºC occurred during winter months December – February, although these results are less conclusive as there is a greater degree of inter-decadal variation. This value is almost twice the global mean from 1904 -2005 of 0.74ºC. This is consistent with above global average regional temperature increases elsewhere in the Central Asian region (Christensen *et al.*

Precipitation trends were far from conclusive as is characteristic of the naturally high variability of precipitation in comparison with temperature change. However, autumn months of September – November displayed a significant positive trend in precipitation levels, although as autumn only contributes approximately 11% to annual precipitation, an increasing trend will only have modest influence on the annual water budget. A possibly more important trend, due to its magnitude is the increase in spring precipitation. This trend wasn't statistically significant but close inspection showed that the past 5 decades have all been above the 100 year mean whereas 4 out of 5 decades prior to that had been below the mean. These results are in keeping with an observed regional trend of increasing precipitation in Central Asia (Trenberth *et al.* 2007, Aizen 2007), possibly related to the remote forcing of cyclical ENSO/NAO events on precipitation over the whole of Central and South – West Asia. This demonstrates the inter-annual variation in precipitation that is characteristic of the region (Syed *et al.* 2005)

Snow cover extent during spring is negatively correlated with temperature (significant). No such correlation was found between winter snow cover and temperature largely because conditions are much less marginal in winter. Spring snow cover is closer to melt point over greater areas therefore small increases in temperature have a greater effect in spring than winter. This is consistent with observed global trends (Brown 2000, Dye 2002, Stewart 2004).

Snow cover extent in spring was positively correlated with spring precipitation. No correlation was found either between winter snow cover and precipitation or perhaps more surprisingly winter precipitation and spring snow cover. The lack of winter correlation is due to the fact that during the core winter months snow cover is generally at a maximum as all precipitation greater than 2000masl is most likely to fall as snow, additionally thaws are unlikely – thus, the result is fairly uniform snow cover results. However variable precipitation rates will effect the depth of snow pack but this impact is generally undetectable when studying snow cover extent until spring when more marginal; conditions differentiate between shallow and deep snow packs, as snow depletion curves reveal.

The extent of spring snow cover (March – May) reduced on average by 15% between 1972-81 and 1998- 07. Indications were found that winter snow cover has decreased by 4%, but these findings were not significant. This is consistent with the negative correlation found between spring snow cover and

temperature together with observed increasing temperature trend. This is confirmed by Haritashya *et al.* (2006) who observed an increase of the snowline in the central Hindu Kush by 200m over the last 30 years. It is therefore reasonable to suggest that increasing spring temperatures in the upper watershed areas of the Kunduz river basin over the course of the 20th century has contributed to a decrease in the extent of spring snow cover – at least over the observed period 1972-2007.

A decadal mean for 1998-2007 revealed that snow cover extent has been below the long term average in all months. The minimum anomaly was in January with increasing deviation below the mean to a maximum anomaly in May. This follows a trend of increasingly marginal snow conditions as spring progresses amplifying the effect of increased temperatures on snow cover and therefore signalling a contraction of the snow season

Total basin peak discharge appears to be positively correlated with the extent of spring snow cover. The observed reduction in spring snow cover extent is therefore expected to lead to an equivalent reduction in discharge resulting from snowmelt. However lack of river discharge data since the 1970s do not allow for this to be verified on the ground.

A significant correlation between basin discharge and the area of irrigated wheat suggests that decreased discharge results in a decreased area planted to wheat. This is a critical result as irrigated wheat represents 87% of the total wheat crop in the KRB and is a staple crop of the Afghan diet (FAO 2003). However available data only spans 8 years and under these limitations conclusions should be drawn cautiously. Additionally, there are many other factors influencing the area of wheat planted and more importantly the yield from a given area, which shall be discussed in the following section. However, the fact that basin hydrology is limiting on irrigated wheat in the KRB is also well illustrated by the construction of snow depletion curves which provide a proxy for snow pack depth and therefore water equivalent. Qualitative inspection of these curves demonstrates a dependence of irrigated agriculture on the snow pack.

6.2. Impact of Climatic Trends on Water Resources

Snow pack and runoff

Linear regression analysis of observed snow cover - temperature/precipitation relationships allowed a quantitative projection of the relative impacts of future changing regimes of temperature, precipitation based on current empirical data and IPCC SRES emission scenarios (Nakicenovic and Swart 2000). It was found that likely temperature change would have a far greater effect on snow pack than the likely changes in precipitation regimes. Even a possible increase in precipitation could not make up for losses as a result of increased temperature. Additionally the precipitation projection assumes that the same proportion of precipitation falls as snow as it does currently – which would not be the case under increasing temperatures.

Snow cover is likely to continue to contract at its margins both in terms of altitude and season as a result of increasing temperature and an increasing proportion of precipitation falling as rain. This will reduce the snow pack resulting in reduced summer water resources and place ever greater demands on the equitable use of water amongst farmers and provide greater challenges for irrigation systems which must often convey water large distances.

The observed trend of contraction of snow season is likely to result in shifts in timing of peak runoff due to an earlier melting snowpack (Nohara *et al*. 2006). Additionally there is the risk that a reduced snow pack would be less likely to support perennial flow of rivers. While no river discharge data is available currently, river levels have been monitored (pers. Comm. Jelle Beekma) as these relate to discharge rates. Interestingly, there appears to be no noticeable difference in the peak water levels over the past 33 years, being end June/beginning July in the early 1970s as well as now (pers. Comm Jelle Beekma 2007). Clearly this is an area that needs careful monitoring as shifts in peak runoff will pose a threat to cropping patterns and reduce availability of water in the peak growing season. However, there are most likely to be shifts in growing season timing with warming. The extent to which cropping patterns can be accommodated in to a changing climate regime will most likely be the key challenge.

Floods and droughts

With the observed warming trend in the KRB both the moisture holding capacity of the atmosphere and evaporation into the atmosphere increase, causing increased climatic variability and an acceleration of the hydrological cycle (Huntington 2006). The end result of this is not necessarily a change in absolute precipitation amounts but rather a change in the characteristics of precipitation with more intense precipitation and conversely more droughts (Trenberth *et al.* 2003).

A higher proportion of winter precipitation falling as rain will lead to increased direct runoff. Increased direct runoff will be augmented by earlier and more rapid snowmelt with higher temperatures increasing the flood risk in winter and spring. Floods are extremely destructive to livelihoods in the

Kunduz basin destroying irrigation infrastructure, agriculture and livestock, especially where people have little adaptive capacity.

As summer precipitation is extremely limited in the Kunduz basin, increased temperature is likely to have the greater direct effect on summer water resources through greater evapotranspiration losses. These losses would be caused by direct evaporation of surface water, increased use of irrigation water and greater transpiration losses of groundwater by vegetation (Huntington 2006). This effect will be felt most acutely in lowland regions of the basin where there is already a negative evapotranspiration/ precipitation balance (eV > p) such as the lowlands (Anderson 2006).

As a largely semi arid country drought has often been a feature of climatic variability in Afghanistan (Bhattacharyya *et al*. 2004), however drought events are likely to become more frequent and intensify as a result of increased temperatures and reduced snowpack (Barnett *et al*. 2005. Indeed the recent severe drought demonstrates the catastrophic effects of low winter snowfall and anomalously high temperatures in the KRB. Additionally a regression analysis puts the likely range of mean spring snow cover by the end of the 21st century at similar levels to that of 2001 with a likely temperature rise of 3.5-4.5ºC. Is this recent drought perhaps an insight into future climatic scenarios?

Groundwater

Groundwater is more problematic to assess as a lack of reliable data makes it difficult to monitor any change as a result of climatic forcing (Alim 2006). Reduced summer precipitation in upper catchment areas (lowland precipitation is negligible) combined with greater transpiration losses are likely to have a detrimental effect on groundwater resources (Eckhardt and Ulbrich 2004). Groundwater levels have been found to correlate more strongly with precipitation than temperature as it doesn't tend to be effected by evaporation to the same extent as surface flows (Chen et al. 2004). Therefore, possible increases in winter precipitation and direct runoff due to winter rains and flood events will play an increasing role in recharge over snowmelt.

Other considerations are that increasing population and economic development will place greater demands on this resource. (Qureshi 2002). This is especially the case if increased ground water extraction is needed to offset reduced surface water flows for agriculture, domestic or industrial use. There has been much evidence of falling water tables with wells running dry during the recent drought, suggesting Afghanistan's groundwater is a relatively fragile resource. Furthermore, demand is projected to increase from its current level of 3BCM (Billion Cubic Metres) to 10BCM in the next 10 years (Quereshi 2002).

Water quality

With warmer temperatures water quality deteriorates by reducing the amount of oxygen that can be dissolved and used for biodegradation. This can lead to an increase in water borne vectors and spread of disease (Kundzewicz *et al*. 2007). Sanitation is generally insufficient in urban centres in the KRB

and contamination of ground water by faecal coliforms as well as agricultural chemicals has been recorded (Riedmann and Beekma 2005). Warmer temperatures are likely to combine with reduced flow rates of water courses in summer leading to more standing water and associated disease. Conversely, extreme precipitation events in winter and spring associated with enhanced hydrological cycle can result in more nutrients, toxins and pathogens being washed into water sources (Chang *et al.* 2001).

Changing patterns of demand

It is important to consider non-climatic drivers in a vulnerability analysis of climate change impacts. The most important of these is likely to be population growth and the associated increase in demand for water. Additionally, one of Afghanistan's top development priorities is food security which entails an intensification and extensification of agriculture and inevitably increased water consumption. There are also signs of growth in the industrial sector in the KRB which will inevitably place greater demands on water resource (Riedmann and Beekma 2005). Conversely, economic growth can lead to more efficient water use with enhanced technology in both agricultural and industrial use although there is generally a time lag between industrialisation and 'greening' of industry. Poverty and population growth together with lack of adaptive capacity, technology and education are cross cutting issues that enhance all these impacts and determine the ability to cope with climatic stresses.

6.3. Mitigation of Climate Change: Water Resource Management

Adaptation and mitigation strategies are likely to be important tools for the inhabitants of the KRB to cope with climatic stresses on water resources. This is achieved through a focus on the non-climatic drivers which affect available water resources and may generally be termed integrated water resource management because of its appreciation of physical, environmental and social processes that must be addressed to ensure the sustainable use of water resources.

Water Policy and Management Structures

Coherent government policy is a crucial factor in effective water management. Due to the transitional status of Afghanistan, the government is often inefficient with several ministries involved with water resource management but lacking both coordination and resources (Beekma and Foxwell 2005, Qureshi 2002). An overarching national water authority would enable the development of a national water resource management strategy as well as the co-ordination of other ministries in its implementation and achievement.

Economic policy can play a vital role in water conservation through placing value upon it as an economic (limited) good; for example water pricing (pers. comm. Sultan Mahmoud, Director of Department of Water Management, Kabul). Other policy decisions that would impact heavily on water resources are likely to be made in the agricultural sector - as this is the main water consumer in the KRB. For example, decisions to import agricultural products imply import of water in a 'virtual' form thereby conserving domestic supplies. In this way Afghanistan's water resources have been subsidised for the last three decades as it has relied on imports and food aid since before the Soviet invasion in 1979.

Monitoring systems play a crucial role in reporting possibilities of floods or droughts due to variation in climatic conditions. These systems require the rehabilitation of destroyed and looted meteorological and hydrological stations as well as the application of satellite data. The international community currently fills this role to some extent but there is a scarcity of ground-truthing (eg. USGS Famine Early Warning System, http://www.fews.net/).

The government has a role to play in agreements over transboundary waters. The North East of Afghanistan provides a significant part of the headwaters of the Amu Darya River, one of the two of the great Central Asian Rivers that feed the Aral Sea. As a consequence of rapid agricultural expansion (through irrigation) in the former Soviet Union, the Aral Sea has become one of the biggest environmental disasters on earth; shrinking to 10% of its former volume over the last 40 years (Micklin 2007). Since the early 1990's when the Central Asian Republics gained independence, a transboundary water agreement was signed in order to address this issue. It is recognised that Afghanistan will need to consider this international dimension in future policy decision over water allocation in the North East of Afghanistan – including the KRB (Nakayama 2003). The government of Afghanistan has made steps in this direction during 2007 by forming preliminary meeting with neighbours in order to

formulate water sharing treaties. Feasibility studies on major water management structures, such as dams, are planned in order to control flood flows, generate electricity and ensure water resources for the dry season; and by doing so open up more agricultural land (Fipps 2007). This is a sensitive issue as the enhanced use of water resources by Afghanistan, as is most likely with economic development, will necessarily impact on its neighbours (Fipps 2007). Transboundary water treaties are likely to become increasingly important to insure that water does not become a destabilising geopolitical issue for the region (Nakayama 2003).

Management at the local level has a long tradition of community water management implemented by a local Mirab (water master) who is in charge of allocating water resources among different fields in his scheme. The Mirab is supported by a Vakil (lawyer) in dispute settlement (Qureshi 2002). This is a centuries old system but recent conflict and emigration has led to land abandonment and a breakdown of the social structure these systems were often based on. Social development and enhanced security are important for these systems to come back on line. An important consideration here is that community systems aren't distorted by the presence and activities of the international community (Beekma and Foxwell 2005).

Irrigation systems and efficiency of water allocation

As the largest consumer of surface flows the operational efficiencies of irrigated agriculture provide a natural focus for water conservation strategies. Irrigation systems are generally based on century's old techniques and run at efficiencies of around 25-30% (Qureshi 2002, Beekma and Foxwell 2005). Undoubtedly there is room for development through investment in rehabilitation of existing structures as well more efficient technology where appropriate. Clearly poverty is the general caveat in this situation – who would pay for the new equipment? Who would pay for repairs? Additionally, irrigation infrastructure has suffered a great toll as a consequence of the war and lack of maintenance due to farmer migration. More modern equipment has often been looted. Many canals are highly silted as a result of a lack of regular maintenance. It has been estimated that approximately one half of all irrigation schemes in Afghanistan need rehabilitation (Qureshi 2002) – this is currently a major area of donor focus.

Improved on farm management is not only linked to technology but also practice. For example, using the optimum level of water in irrigation is crucial to maintain yields, reduce salinisation of soils due to water-logging and to promote efficiency of water use (Anderson 2006). Control of water intake plays a major role in irrigation flows but this is a considerable challenge due to the wide range of river discharge magnitudes that such systems have to cope with. The traditional strategy is to construct semi-permanent systems from earthen clods and straw which must be reconstructed after flood damage. This requires a large amount of manpower but does not require expensive material inputs (Anderson 2006).

UNEP observed in a recent report (2003) that despite recent drought conditions water conservation measures were not in place. Additionally, there was no use or promotion of crops with low water requirements at this time.

With earlier runoff occurring due to increased temperatures as well as an increasing amount of winter precipitation falling as rain, water harvesting and storage capacity are likely to play an increasingly important role in smoothing the temporal variation in availability of water and ensuring supplies in the dry summer months. As stated above, draft versions of the Goverments transboundary water policy provides provision for major water management structures – however more local systems which are 'low tech' and readily replicable are likely to be of more use to the vast majority of rural communities (Aini 2007).

Watershed management

The concept of watershed management involves an evolution from focusing on a single objective such as conserving soil and water or reversing deforestation – to having multiple objectives, including social and economic development and natural resource conservation (Vania and Taneja 2005).

Upper catchment management is of utmost importance in the KRB as elsewhere in Afghanistan where three decades of conflict, poverty and lack of governance have manifested in extensive environmental degradation in upper watershed areas, including deforestation and overgrazing. Watershed management strategies such as afforestation, soil conservation and pasture management are critical to improve water storage capacity, decrease temporal variability in discharge, increase groundwater recharge and improving livelihoods in remote upland areas. (See chapter 3 for fuller discussion of watershed issues).

The subject of watershed management is critical now given the severe state much of these regions are currently in and its importance will increase with the projected impacts of climate change. The key question, however, is how do you get those who are often the poorest in marginal upland regions to conserve resources for the benefit of, often richer, down stream users? Market based compensation systems have been shown to work in many parts of the world whereby upstream communities are paid for watershed conservation services by downstream beneficiaries (Rosa *et al.* 2004, Bond 2007, Porras et al. 2007). However, these schemes are by no means straightforward and are multi-dimensional in implementation; therefore success is heavily dependent on the local context. These systems can work at either a local grass roots scale or as part of a broader government policy. Emphasis must be placed on the interconnectivity of the river basin system through education so as to create a 'willingness-to-pay' in downstream users of watershed services. Such structures are probably ambitious at this stage in development process of the KRB, but could play a future role.

Coupled with this point is the fact that poverty will often promote unsustainable practices through the necessity of day-to-day survival. Therefore, poverty reduction and improvement of livelihoods in upland marginal areas can increase the communities capacity as effective resource managers.

7. CONCLUSION AND RECOMMENDATIONS

Observed warming trends in both winter and spring, has resulted in reduced snow cover in the upper watershed regions of the Kunduz River Basin. This reduction has been detected over the past decade in all months (December – May) but is greatest at the end of spring, therefore signalling a contraction of the snow season. Possible increase in winter precipitation is unlikely to compensate for the reduction of spring snow cover particularly at marginal elevations where it is increasingly going to fall as rain. This trend is likely to continue with projected warming trends in the 21st Century altering the quantity and timing of surface flows.

Reduced snow cover together with earlier spring melt is likely to have serious implications for irrigated agriculture, which is the main economic basis of the Kunduz River Basin. The frequency of spring floods and summer droughts is likely to increase as a result of more intense precipitation events combined with earlier, more rapid snowmelt and higher evaporation rates; all being driven by increasing temperatures. These natural disasters place heavy tolls on livelihoods and communities in the KRB and are likely to continue to do so with increasing frequency.

Groundwater resources are likely to be exploited to an increasing extent in order to compensate for reduced surface flows, particularly in summer. It is crucial that these resources are accurately surveyed and monitored to prevent over abstraction. Careful waste management is also needed to prevent contamination of these resources.

The recent drought at the turn of the millennium is possibly an insight into future climatic conditions with reduced snowpack and enhanced temperature combining to create widespread drought conditions. To what extent possible increase in winter rains will mitigate these conditions is unclear.

Rapidly increasing population levels, agricultural expansion and economic development are likely to increase future demand for water resources. Careful water management will be needed to ensure these demands are met.

However there is a good deal of room for mitigation of these future scenarios via many other factors which influence water use efficiency. Key areas of focus are government institutions and policy, on farm management, rehabilitation of irrigation infrastructure and watershed protection. As Afghanistan emerges from its current sate of instability its likely to have an increasing role in transboundary water agreements with other Central Asian countries concerning the Amu Darya River basin, of which the KRB is a major component. This may have implications for future water use as well as agricultural policy in general.

The security situation is a cross cutting issue which negatively affects economic development, government process and rebuilding of the country. While there is much international focus currently on development in Afghanistan, stability is a precondition for this focus to be effective. Finally,

endemic poverty levels pose a serious threat to the capacity of the people of Afghanistan to cope with the negative impacts of climatic change and will most likely pose the greatest challenge in the near future.

Monitoring

- An accurate assessment of water resources in the KRB desperately requires the reconstruction of meteorological observation stations and hydrological gauging stations. Baseline data on surface hydrology and baseflows still appears to be absent. While satellite data can provide a good amount of information, ground truthing is essential to validate data and ensure reliability of early warning systems.

- River discharge data is sorely needed to allow for accurate monitoring of shifts in quantity or seasonality of peak river discharge.

- Groundwater is likely to play an increasing role in future water use and therefore surveying of available resources is an important task. Ensuring the purity of this resource will be a major challenge with agricultural expansion and population growth. Waste management is likely to be a key issue here.

- Future work could benefit from undertaking an analysis of snow derived water resources at the sub basin level which is possible with the finer resolution that the MODIS sensor provides since 2002. This would enable a greater understanding of hydrological differences that exist within the basin.

Adaptation and Mitigation

- Continuing watershed management activities are a key component of mitigating climate change impacts and regulating flows as well as maintaining water quality. Payment for environmental services may provide incentive for upper catchment communities to manage resources sustainably. However, currently, the necessary institutional arrangements for this are largely absent in the KRB and Afghanistan in general.

- Promoting water efficiency through improved agricultural practice, rehabilitation of irrigation systems and possible introduction of irrigation control technologies is likely to deliver significant water savings.

- The government has an important role in developing a coordinated water sector. Major water management structures such as dams are likely to play a large role in controlling flood flows and providing water resources in the dry season.

REFERENCES

Agrawala, S., Barlow, M., Cullen, H. and Lyon, B. (2001). *The Drought and Humanitarian Crisis in Central and Southwest Asia: A Climate Perspective*. The International Research Institute for Climate Prediction (IRI), New York,

Aini, A. (2007). Water Conservation in Afghanistan. *Journal of Developments in Sustainable Agriculture*: 2: 51-58.

Aizen, V.B., Aizen, E.M. (2007). Geo-informational simulation of possible changes n Central Asian water resources. *Global and Planetary Change* **56**: 341–358.

Aizen VB, Aizen EM, Melack JM, and Dozier J. (1997). Climatic and hydrologic changes in the Tien Shan, Central Asia. *Journal of Climate*, **10** (6): 1393-1404

Alim, A.K. (200) Present Situation of the Sustainable Management of Water Resources for Human Survival and Bio-production, and the Role of Agro-environmental Education Today in Afghanistan.

Alim, A.K. (2006). Sustainability of Water Resources in Afghanistan. *Journal of Developments in Sustainable Agriculture* 1:53-66.

Alim, A.K. and Shobair, S.S. (2002). *Drought and human suffering in Afghanistan*. Country paper presented at TASAE2002, Japan.

ANDS (2005). Afghanistans National Development Strategy.
http://www.ands.gov.af/

Anderson, I.M. (2006). *Water Management, Livestock and the Opium Economy*. Afghanistan Research and Evaluation Unit. Kabul

Anisimov O.A, Nelson F.E. (1996). Permafrost distribution in the Northern Hemisphere under scenarios of climatic change. *Global and Planetary Change*, **14** (1-2): 59-72.

Armstrong, R. L., and M. J. Brodzik. 2005. *Northern Hemisphere EASE-Grid weekly snow cover and sea ice extent version 3*. Boulder, Colorado USA: National Snow and Ice Data Center. Digital media. (http://nsidc.org)

Archer, D. (2003). Contrasting hydrological regimes in the upper Indus Basin. *Journal of Hydrology*, **274:** 198–210.

Arnell, N.W. (2004). Climate change and global water resources: SRES emissions and socio-economic scenarios. *Global Environmental Change-Human and Policy Dimensions* **14**(1): 31-52

Arnell, N.W. (2003). Effects of IPCC SRES emissions scenarios on river runoff: a global perspective *Hydrology and Earth System Sciences*, **7**(5): 619-641

Arnell, N.W. (1999). Climate change and global water resources. *Global Environmental Change,* **9**: S31- S49

Bhattacharyya, K.; Azizi, P. M.; Shobair, S. S.; Mohsini, M. Y. (2004). *Drought impacts and potential for their mitigation in southern and western Afghanistan.* Working Paper 91. Colombo, Sri Lanka: International Water Management Institute.

Bamzai, A.S. (2003). Relationship between snow cover variability and arctic oscillation index on a hierarchy of time scales. *International Journal of Climatology,* **23**: 131–142.

Barlow, M., Cullen, H., Lyon, B. (2002). Drought in central and southwest Asia: La Nina, the warm pool, and Indian Ocean precipitation. *Journal of Climate*, **15**(7): 697-700.

Barnett, T.P., Adam, J.C., and Lettenmeier, D.P. (2005). Potential impacts of a warming climate on water availability in snow-dominated regions. *Nature*, 438: 303–309.

Barnett, T., Malone, R., Pennell, W., Stammer, D., Semtner, B., and Washington, W. (2004). The effects of climate change on water resources in the west: Introduction and overview. *Climatic Change*, **62**(1-3): 1-11

Barry, R.G. (1990). Changes in Mountain Climate and Glacio-Hydrological Responses. *Mountain Research and Development*, **10**(2): 161-170.

Beekma, J. and Foxwell, S. (2005). Kunduz River Basin Programme: TA to the Ministry of Irrigation, Water Resources and Environment for the Implementation of the Food Security/Water Management Project in Kunduz, Baghlan and Takhar Provinces, Inception Report, Landell Mills, Trowbridge.

Beniston M. (2003). Climatic Change in Mountain Regions: A Review of Possible Impacts. Climatic Change, **59**(1-2): 5-31.

Beniston M., Diaz H.F., and Bradley R.S. (1997). Climatic Change at High Elevation Sites: An Overview. Climatic Change, 36(3-4): 233-251.

Beniston, M., and Rebetez, M. (1996). Regional Behavior of Minimum Temperatures in Switzerland for the Period 1979-1993. *Theoretical and Applied Climatology*, **53**: 231-243.

Bond, I. (2007). *Payments for watershed services: opportunities and realities.* IIED

Brown R.D. (2000). Northern hemisphere snow cover variability and change, 1915-97. *Journal of Climate*, **13**(13): 2339-2355.

Chalise, S.R., Kansakar, S.R., Rees, G., Croker, K. and Zaidman, M. (2003). Management of water resources and low flow estimation for the Himalayan basins of Nepal. *Journal of Hydrology,* **282:** 25–35.

Chang, H., B. Evans and D. Easterling. (2001). The effects of climate change on stream flow and nutrient loading. *Journal American Water Resources Association*, **3:** 973-985.

Chen, Z., S. Grasby and K. Osadetz, (2004). Relation between climate variability and groundwater levels in the upper carbonate aquifer, southern Manitoba, Canada. *Journal of Hydrology*, **290**: 43-62.

Chen, M., P. Xie, and J.E. Janowiak, (2002). Global land precipitation: a 50-yr monthly analysis based on gauge observations. *Journal of Hydrometeorology*, **3**, 249–266.

Christensen, J.H., B. Hewitson, A. Busuioc, A. Chen, X. Gao, I. Held, R. Jones, R.K. Kolli, W.-T. Kwon, R. Laprise, V. Magaña Rueda, L. Mearns, C.G. Menéndez, J. Räisänen, A. Rinke, A. Sarr and P. Whetton, 2007: Regional Climate Projections. In: Climate Change 2007: The Physical Science Basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change [Solomon, S., D. Qin, M. Manning, Z. Chen, M. Marquis, K.B. Averyt, M. Tignor and H.L. Miller (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.

Cruz, R.V., H. Harasawa, M. Lal, S. Wu, Y. Anokhin, B. Punsalmaa, Y. Honda, M. Jafari, C. Li and N. Huu Ninh, 2007: Asia. Climate Change 2007: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change, M.L. Parry, O.F. Canziani, J.P. Palutik of, P.J. van der Linden and C.E. Hanson, Eds., Cambridge University Press, Cambridge, UK, 469-506.

Dahe, Q. Shiyin,. L. and Peiji L. (2006). Snow Cover Distribution, Variability, and Response to Climate Change in Western China. *Journal Of Climate*, **19:** 1820-1833.

Dankers, R, And Christensen, O.B. (2005). Climate Change Impact On Snow coverage, Evaporation and River Discharge in the Sub-Arctic Tana Basin, Northern Fennoscandia. *Climatic Change,* **69**: 367–392.

Dash, S. K., Singh, G. P., Shekhar M. S. and Vernekar A. D. (2005). Response of the Indian summer monsoon circulation and rainfall to seasonal snow depth anomaly over Eurasia. *Climate Dynamics* **24**: 1–10.

Dijk, A. and Keenan, R.J. (2007). Planted forests and water in perspective. *Forest Ecology and Management* **251** (1-2): 1-9.

Dore, M.H. (2005). Climate change and changes in global precipitation patterns: What do we know? *Environment International,* **31**: 1167 – 1181.

Dye, D.G. (2002). Variability and trends in the annual snow-cover cycle in Northern Hemisphere land areas, 1972-2000. *Hydrological Processes* **16** (15): 3065-3077

Eckhardt, K. and Ulbrich, U. (2003). Potential impacts of climate change on groundwater recharge and streamflow in a central European low mountain range. *Journal of Hydrology,* **284**: 244–252.

FAO/WFP. (2005). crop and food supply assessment mission to Afghanistan. Field document, Rome.

FAO/WFP. (2004). Crop and food supply assessment mission to Afghanistan. Field document, Rome.

FAO/WFP. (2003). Crop and food supply assessment mission to Afghanistan. Field document, Rome.

FAO/WFP. (2002). Crop and food supply assessment mission to Afghanistan. Field document, Rome.

FAO/WFP. (2001). Crop and food supply assessment mission to Afghanistan. Field document, Rome.

FAO/WFP. (2000). Crop and food supply assessment mission to Afghanistan. Field document, Rome.

FAO/WFP. (1999). Crop and food supply assessment mission to Afghanistan. Field document, Rome.

FAO/WFP. (1998). Crop and food supply assessment mission to Afghanistan. Field document, Rome.

FAO/WFP. (1997). Crop and food supply assessment mission to Afghanistan. Field document, Rome.

Fipps, G. (2007). *Transboundary Water Policy of Afghanistan* (Draft). Kabul.

Gerrard, A.J. (1990). *Mountain Environments*. Belhaven Press,London.

Giorgi F, Hurrell J.W, Marinucci M.R, Beniston M. (1997). Elevation dependency of the surface climate change signal: A model study. *Journal of Climate*, **10**(2): 288-296.

Government of Afghanistan. (2007). Transboundary water policy: Draft

Government of Balochistan/ IUCN Pakistan (2000). *Balochistan Conservation Strategy IUCN Pakistan and GoB*. Karachi, Pakistan.

Graham, L.P, Hagemann, S., Jaun, S., Beniston, M. (2007). On interpreting hydrological change from regional climate models. *Climatic Change* **81**:97–122.

Haritashya, U. K.; Bishop, M. P.; Shroder, J. F.; Bulley, H. N.; Olsenholler, J. A.; Sartan, J. N. (2006). *Snowline and Glacier Assessment in the Central Hindu Kush, Afghanistan, Utilizing Multi- Temporal Satellite Imagery.* American Geophysical Union, Fall Meeting 2006.

Harris, C. Muhll, D.V, Isaksen, K., Haeberli, W., Sollid, J.L., King, L., Holmlund, P. Dramis, F., Guglielmin, M. and Palacios, D. (2003). Warming permafrost in European mountains. *Global and Planetary Change* **39**: 215–225.

Huntington, T.G. (2006). Evidence for intensification of the global water cycle: Review and synthesis *Journal of Hydrology,* **319**(1-4): 83-95.

Iyngararasan, M., Tianchi, L. Shrestha, S., Mool, P.K., Yoshino, M. and Watanabe, T. (2004). *The challenges of mountain environments: water, natural resources, hazards, desertification and the implications of climate change.* In Key issues for mountain areas eds. Price, M.F., Jansky, L. and Latsenia, A.A. United Nations University Press, Tokyo.

Immerzeel, W.W. (2007). Historical trends and future predictions of climate variability in the Brahmaputra basin. *The International Journal of Climatology (in press).*

Immerzeel, W.W; Quiroz, R.A. and de Jong S.M. (2005). Understanding precipitation patterns and land use interaction in Tibet using harmonic analysis of SPOT VGT-S10 NDVI time series. *International Journal of Remote Sensing*, **26**(11): 2281–2296.

Jansson, P., Hock, R. and Schneider, T. (2003). The concept of glacier storage: a review. *Journal of Hydrology,* **282:**116–129

Jianchu, X. Shrestha, A., Vaidya, R., Eriksson, M. and Hewitt, K. (2007). *The Melting Himalayas Regional Challenges and Local Impacts of Climate Change on Mountain Ecosystems and Livelihoods.* International Centre for Integrated Mountain Development (ICIMOD) Kathmandu, Nepal.

Jungo, P. and Beniston, M. (2001). Changes in the anomalies of extreme temperature anomalies in the 20th century at Swiss climatological stations located at different latitudes and altitudes. *Theoretical and Applied Climatology*, **69:** 1-12.

Kelly, A.T. (2003). Rebuilding Afghanistan's Agriculture Sector. Asian Development Bank, Manilla.

Kreutzmann, H. (2001). Development Indicators for Mountain Regions
Mountain Research and Development, **21**(2): 132-139.

Kulkarni, A.V., Bahuguna1, I.M., Rathore1, B.P., Singh, S.K., Randhawa, S.S., Sood, R.K. and Dhar, S. (2007). Glacial retreat in Himalaya using Indian Remote Sensing satellite data. *Current Science*, **92**(1): 10.

Kundzewicz, Z.W., L.J. Mata, N.W. Arnell, P. Döll, P. Kabat, B. Jiménez, K.A. Miller, T. Oki, Z. Sen and I.A. Shiklomanov, 2007: Freshwater resources and their management. Climate Change 2007: Impacts, Adaptation and Vulnerability. Contribution of Working Group II to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change, M.L. Parry, O.F. Canziani, J.P. Palutikof, P.J. van der Linden and C.E. Hanson, Eds., Cambridge University Press, Cambridge, UK, 173-210.

Lang H, Kaser G. (2003). Special issue: Mountain hydrology and water resources – Preface. *Journal of Hydrology,* **282**(1-4): 1-1.

Lemke, P., J. Ren, R.B. Alley, I. Allison, J. Carrasco, G. Flato, Y. Fujii, G. Kaser, P. Mote, R.H. Thomas and T. Zhang, 2007: Observations: Changes in Snow, Ice and Frozen Ground. In: Climate Change 2007: The Physical Science Basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change [Solomon, S., D. Qin, M. Manning, Z. Chen, M. Marquis, K.B. Averyt, M. Tignor and H.L. Miller (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.

Lenton R. (2004). Water and climate variability: development impacts and coping strategies. *Water Science and Technology,* **49**(7): 17-24.

Lioubimtseva E, Cole R. (2006). Uncertainties of climate change in arid environments of Central Asia Reviews in Fisheries Science, **14** (1-2): 29-49.

Marchenko, S.S., Gorbunov, A.P., and Romanovsky, V.E. (2007). Permafrost warming in the Tien Shan Mountains, Central Asia. *Global and Planetary Change,* **56**: 311–327.

Meehl, G.A., T.F. Stocker, W.D. Collins, P. Friedlingstein, A.T. Gaye, J.M. Gregory, A. Kitoh, R. Knutti, J.M. Murphy, A. Noda, S.C.B. Raper,I.G. Watterson, A.J. Weaver and Z.-C. Zhao, 2007: Global Climate Projections. In: *Climate Change 2007: The Physical Science Basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change* [Solomon, S.,D. Qin, M. Manning, Z. Chen, M. Marquis, K.B. Averyt, M. Tignor and H.L. Miller (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.

Mercy Corps (2005). Catchment Development Programme Upstream Water Catchments' Protection and
Forest Regeneration in North Afghanistan (Kunduz River Basin Programme) EUROPEAID/121761/C/G/AF.

Micklin P. (2007). The Aral Sea disaster. Annual Review of Earth And Planetary Sciences, **35**:47-72

Ministry of Agriculture, Irrigation and Livestock. (2007). Agriculture Prospects Report, Kabul

Ministry of Rural Rehabilitation and Development/ World Food Programme. (2006). Afghanistan Food Security Monitoring Bulletin (AFSMB). Kabul/Rome.

Ministry of Water & Power, Afghanistan (2004) *Final Report Project AFG 03170 Power Sector Master Plan Annex C6 – Hydrology*. Kabul.

Mitchell, T.D., and P.D. Jones, 2005: An improved method of constructing a database of monthly climate observations and associated high-resolution grids. *Int. J. Climatol.*, **25**, 693–712

Mitchell,T.D., Hulme,M., and New,M., 2002: Climate data for political areas. *Area* **34**:109-112.

Mountain Forum. (2007). Melting Mountains. *Mountain Forum Bulletin*, **8**: 2
www.mtnforum.org

Nakicenovic N. and Swart, R. (2000). *Special Report on Emissions Scenarios*. IPCC.

Nakayama, M. (2003). *Implications of virtual water concept on management of international water systems: Cases of two Asian international river basins*. *In*: Proceedings of the International Expert Meeting on Virtual Water Trade. Ed A.Y. Hoekstra. IHE Delft.

Nazemosadat, M. J., Samani, N. Barry, D. A., Niko, M. Molaii. (2006). ENSO forcing on climate change in Iran: Precipitation analysis. *Iranian Journal of Science and Technology Transaction B-Engineering* 30 (B4): 555-565.

Nijssen, B., O'donnell, G., Hamlet A.F., and Lettenmaier, D.P. (2001). Hydrologic sensitivity of global rivers to climate change. *Climatic Change* **50**: 143–175,

Nohara, D., Kitoh, A, Hosaka, M., Oki, T. (2006) Impact of climate change on river discharge projected by multimodel ensemble. *Journal of Hydrometeorology* **7**(5): 1076-1089.

Qian B and Saunders M.A. (2003). Summer U.K. temperature and its links to preceding Eurasian snow cover, North Atlantic SSTs, and the NAO. Journal of Climate, **16**: 4108-4120.

Ososkova T, Gorelkin N, Chub, V. (2000). Water resources of Central Asia and adaptation measures for climate change. *Environmental Monitoring and Assessment,* **61**(1): 161-166

Porras, I.,Grieg-Gran, M. and N. Neves. (2007) All That Glitters: A review of payments for watershed services in developing countries. IIED.

Qureshi, A.S. (2002). Water Resources Management in Afghanistan: The Issues and Options. Working Paper 49 Pakistan Country Series No. 14. IWMI.

Ragab, R. and Prudhomme, C. (2002). Climate Change and Water Resources Management in Arid and Semi-arid Regions: Prospective and Challenges for the 21st Century. *Biosystems Engineering*, **81**(1), 3-34.

Randall, D.A., R.A. Wood, S. Bony, R. Colman, T. Fichefet, J. Fyfe, V. Kattsov, A. Pitman, J. Shukla, J. Srinivasan, R.J. Stouffer, A. Sumi and K.E. Taylor, 2007: Climate Models and Their Evaluation. In: Climate Change 2007: The Physical Science Basis. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change [Solomon, S., D. Qin, M. Manning, Z. Chen, M. Marquis, K.B. Averyt, M.Tignor and H.L. Miller (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.

Reuben Paul, P., Ramana, L.V. and Sankar, S. (1994). Estimation of basin Snow water equivalent (SWE) using accumulation and depletion patterns of Snowcover from optical satellite Data. *GISdevelopment.*

Riedmann, F. and Beekma, J. (2005). *Implementing the Kunduz River Basin Authority*, Paper presented at the Kabul Water Conference.

Rosa, H, S. Kandel and L. Dimas, (2004). Compensation for environmental services and rural communities: lessons from the Americas, *International Forestry Review* **6** (2): 187–194

Seidel, K., Ehrler C. and Martinec, J. (1998). Effects of climate change on water resources and run off in an Alpine basin. Hydrological Processes, **12**: 1659-1669.

Shaman, J. and Tziperman, E. The Effect of ENSO on Tibetan Plateau Snow Depth: A Stationary Wave Teleconnection Mechanism and Implications for the South Asian Monsoons. Journal of Climate, **18**: 2067 – 2079.

Shamir, E. and Georgakakos, K.P. (2007). Estimating snow depletion curves for American River basins using distributed snow modelling. *Journal of Hydrology* **334** (1-2): 162-173.

Shobair, S.S. (2001). *Current Drought Situation in Afghanistan,* FAO Afghanistan.

Singh, P. Arora, M. and Goel, N. K. (2006). Effect of climate change on runoff of a glacierized Himalayan basin. Hydrological Processes, **20:** 1979–1992.

Singh, P. and Bengtsson, L. (2005). Impact of warmer climate on melt and evaporation for the rainfed, snowfed and glacierfed basins in the Himalayan region. Journal of Hydrology, **300**: 140–154.

Singh, P. and Bengtsson L. (2004). Hydrological sensitivity of a large Himalayan basin to climate change. Hydrological Processes, **18**: 2363–2385.

Singh, P. Bengtsson, L. (2003). Effect of warmer climate on the depletion of snow covered area in the Satluj basin in the western Himalayan region. *Hydrological Sciences–Journal–des Sciences Hydrologiques*, **48**(3): 413-425.

Singh, P. and Kumar, N. (1997). Impact assessment of climate change on the hydrological response of a snow and glacier melt runoff dominated Himalyan river basin. *Journal of Hydrology*, **193**:316 – 350.

Stewart, I.T. Cayan, D.R. and Dettinger, M.D. (2004). Changes in snowmelt runoff timing in Western North America under a 'business as usual' climate change scenario. *Climatic Change* **62**: 217–232.

Syed, F. S., Giorgi, F., Pal, J.S. and King, M.P. (2006). Effect of remote forcings on the winter precipitation of central southwest Asia part 1: observations. *Theoretical and Applied Climatology*. **86**, 147–160.

Trenberth, K.E., P.D. Jones, P. Ambenje, R. Bojariu, D. Easterling, A. Klein Tank, D. Parker, F. Rahimzadeh, J.A. Renwick, M. Rusticucci, B. Soden and P. Zhai, 2007: Observations: Surface and

Atmospheric Climate Change. In: *Climate Change 2007: The Physical Science Basis*. Contribution of Working Group I to the Fourth Assessment Report of the Intergovernmental Panel on Climate Change [Solomon, S., D. Qin, M. Manning, Z. Chen, M. Marquis, K.B. Averyt, M. Tignor and H.L. Miller (eds.)]. Cambridge University Press, Cambridge, United Kingdom and New York, NY, USA.

Trenberth KE, Dai AG, Rasmussen RM, Parsons DB. (2003). The changing character of precipitation. *Bulletin of the American Meteorological Society,* **84**(9): 1205- 1218.

Tsonis, A.A, Hunt, A.G., and Elsner, J.B. (2003). On the relation between ENSO and global climate change. *Meteorology and Atmospheric Physics,* **84**(3-4): 229-242.

Uhl V.W. (2006). Afghanistan: An Overview of Ground Water Resources and Challenges. Ground Water. **44**(5): 626–627

UNEP (2007)

UNEP. (2003). Post-Conflict Assesment: Afghanistan. Geneva.

USAID. (2007). *The Contribution of Regional Markets to Afghan Wheat Supplies*. FEWS NET Afghanistan Washington DC.

USDA. (2003). *Afghanistan: Crop Condition Update 2003/04 Wheat.* Production Estimates and Crop Assessment Division Foreign Agricultural Service. http://www.pecad.fas.usda.gov/

Vania, F. and Taneja, B. **(**2005). *People, Policy, Participation: Making Watershed Management work in India*. IIED.

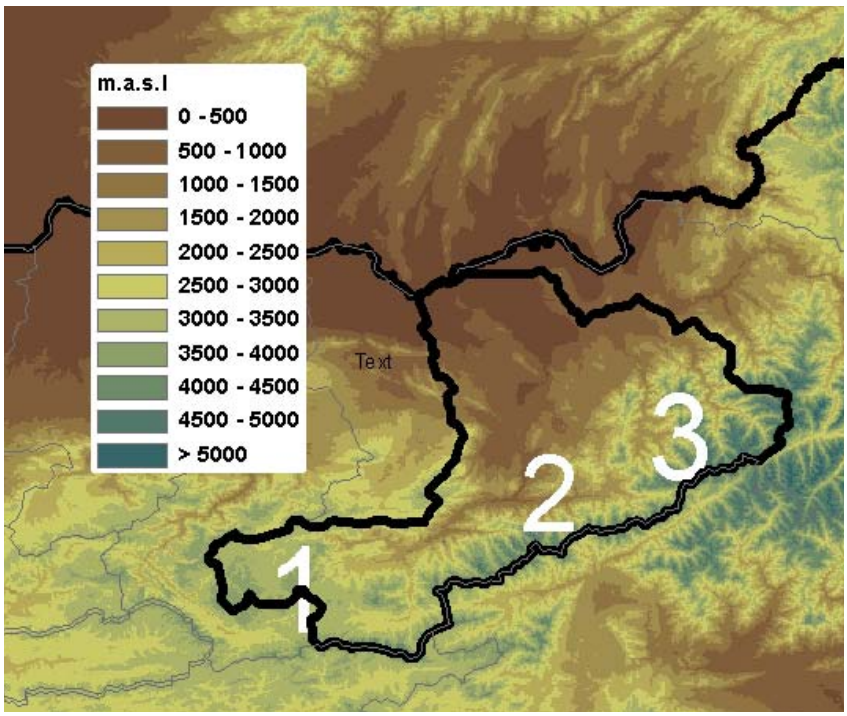
Weingartner, R., Barben, M., and Spreafico, M. (2003). Floods in mountain areas—an overview based on examples from Switzerland. *Journal of Hydrology* **282:** 10–24.

Winiger, M., Gumpert, M. and Yamout, H. (2005). Karakorum–Hindukush–western Himalaya: assessing high-altitude water resources. *Hydrological Processes* **19**: 2329–2338

Xu, C.Y, and Halldin S. (1997). The effect of climate change on river flow and snow cover in the NOPEX area simulated by a simple water balance model. *Nordic Hydrology,* **28**(4-5): 273-282.

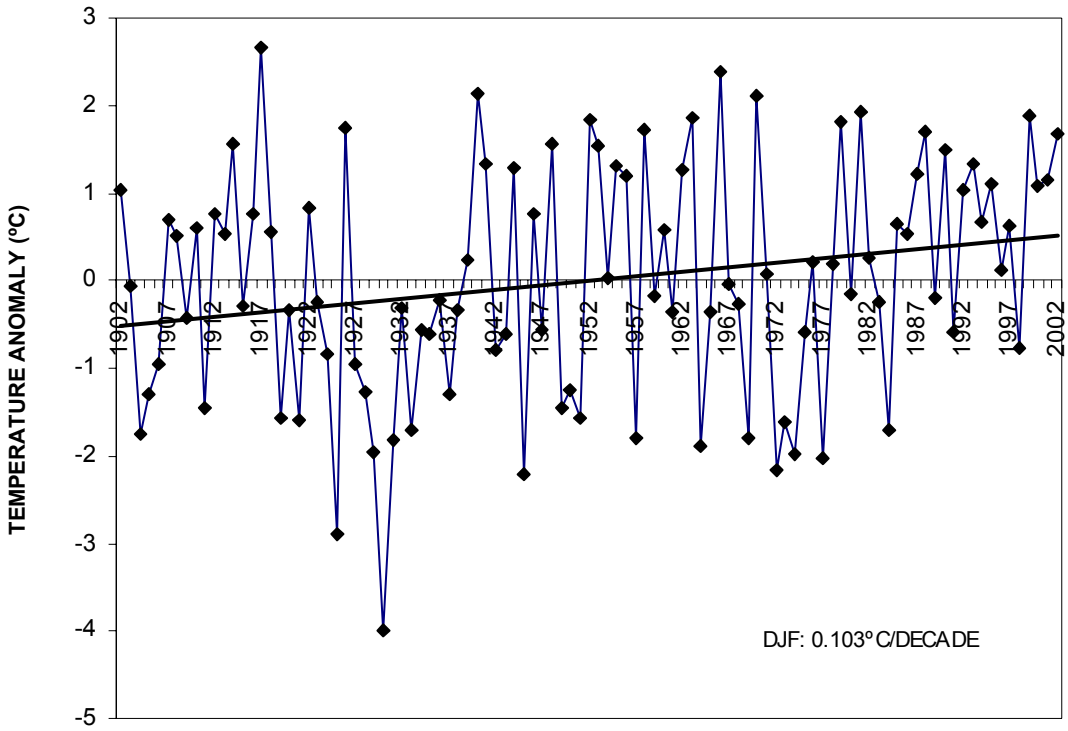
Zhang, Y., Li, T. and Wang, B. (2004). Decadal Change of the Spring Snow Depth over the Tibetan Plateau: The Associated Circulation and Influence on the East Asian Summer Monsoon. *Journal of Climate*, **17:** 2793-2780.

APPENDIX I: CLIMATOLOGY

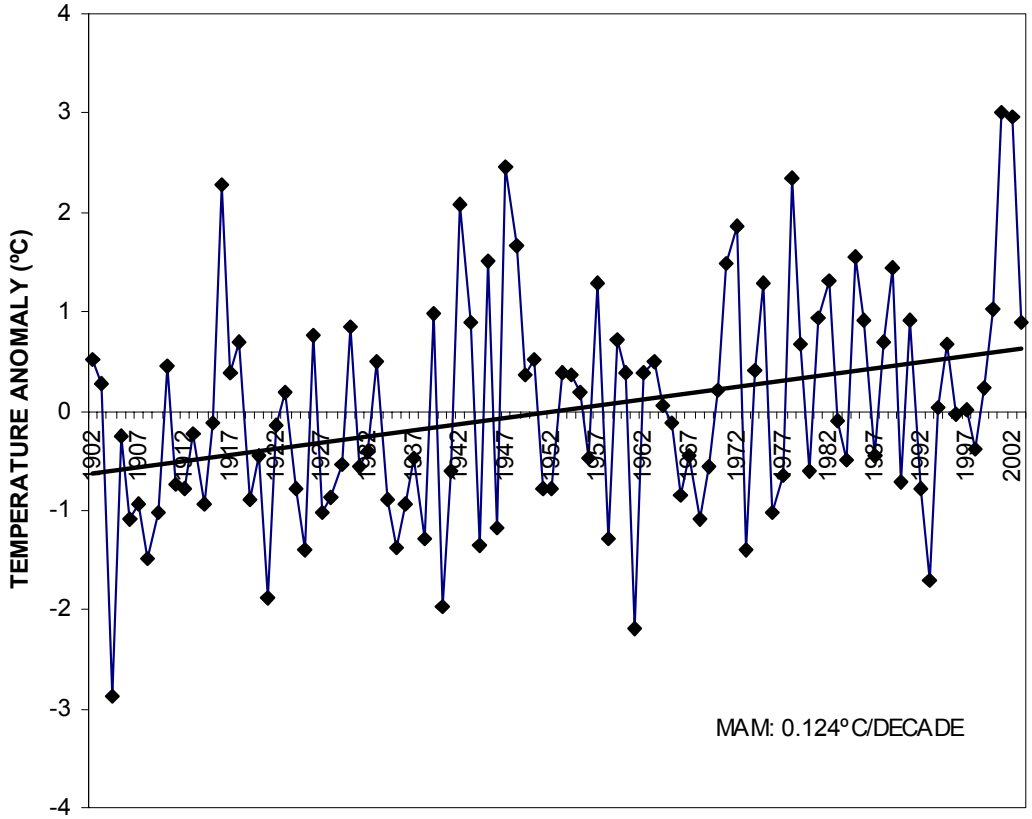


Number	CRU grid	Longitude (centre)	Latitude (centre)	Altitude (masl)
1	495, 250	67.5	35	3368
2	498, 251	68.5	35.5	3347
3	500, 251	70	36	3519

Position of CRU 0.5ºx0.5ºclimate grids used in Temperature and precipitation analysis



Winter (DJF) complete temperature time series 1902-2002.



Spring (MAM) complete temperature time series 1902-2002

CRU temperature dataset (ºC) constructed from mean values of grids 495, 250 /498, 251 /500, 251.

	DEC	JAN	FEB	MAR	APR	MAY	DJF MEAN	DJF ANOMALY	MAM MEAN	MAM ANOMALY
1902	-4.1	-6.0	-6.5	-1.8	2.1	8.7	-5.5	1.0	3.0	0.3
1903	-4.5	-9.5	-5.9	-6.1	0.1	5.5	-6.6	-0.1	-0.2	-2.9
1904	-7.1	-13.3	-4.6	-1.6	1.9	7.0	-8.3	-1.7	2.4	-0.2
1905	-4.2	-9.2	-10.3	-5.3	2.2	7.9	-7.9	-1.3	1.6	-1.1
1906	-5.4	-9.0	-8.1	-2.4	0.3	7.3	-7.5	-0.9	1.7	-0.9
1907	-3.0	-6.3	-8.3	-3.9	3.0	4.5	-5.9	0.7	1.2	-1.5
1908	-5.5	-6.6	-6.0	-3.2	2.8	5.4	-6.1	0.5	1.7	-1.0
1909	-5.2	-9.2	-6.6	-1.3	4.2	6.5	-7.0	-0.4	3.1	0.5
1910	-5.0	-7.4	-5.5	-3.8	1.3	8.3	-6.0	0.6	1.9	-0.7
1911	-10.2	-8.6	-5.3	-3.2	1.7	7.3	-8.0	-1.4	1.9	-0.8
1912	-6.5	-7.5	-3.4	-2.4	2.8	6.9	-5.8	0.8	2.5	-0.2
1913	-5.5	-5.2	-7.4	-4.7	1.7	8.2	-6.0	0.5	1.7	-0.9
1914	-4.3	-3.9	-6.8	-2.6	3.3	7.0	-5.0	1.6	2.6	-0.1
1915	-7.2	-6.3	-7.0	1.8	3.2	9.8	-6.9	-0.3	5.0	2.3
1916	-4.8	-5.0	-7.6	-1.0	4.0	6.3	-5.8	0.8	3.1	0.4
1917	-5.6	-3.8	-2.4	-0.9	3.6	7.5	-3.9	2.7	3.4	0.7
1918	-4.5	-7.4	-6.2	-3.5	0.3	8.6	-6.0	0.6	1.8	-0.9
1919	-9.3	-9.6	-5.6	-2.5	3.3	5.9	-8.1	-1.6	2.2	-0.5
1920	-5.4	-6.4	-8.9	-2.3	-0.2	4.9	-6.9	-0.3	0.8	-1.9
1921	-10.9	-7.6	-5.9	-2.8	4.2	6.2	-8.2	-1.6	2.5	-0.1
1922	-4.5	-7.0	-5.7	-1.0	3.3	6.3	-5.7	0.8	2.9	0.2
1923	-4.8	-8.2	-7.4	-2.1	1.5	6.3	-6.8	-0.2	1.9	-0.8
1924	-8.1	-7.5	-6.6	-2.0	2.2	3.7	-7.4	-0.8	1.3	-1.4
1925	-7.3	-10.7	-10.4	-1.0	3.5	7.8	-9.5	-2.9	3.5	0.8
1926	-3.1	-5.9	-5.4	-2.4	0.7	6.7	-4.8	1.7	1.7	-1.0
1927	-4.8	-10.0	-7.8	-3.6	2.6	6.4	-7.5	-0.9	1.8	-0.9
1928	-5.4	-10.9	-7.2	-5.0	3.4	8.1	-7.8	-1.3	2.1	-0.5
1929	-6.4	-10.8	-8.3	-0.4	4.8	6.3	-8.5	-2.0	3.5	0.8
1930	-11.5	-12.5	-7.7	-2.1	1.5	7.0	-10.6	-4.0	2.1	-0.6
1931	-7.0	-7.6	-10.6	-1.5	3.6	4.7	-8.4	-1.8	2.3	-0.4
1932	-7.1	-7.4	-6.2	-0.8	4.7	5.7	-6.9	-0.3	3.2	0.5
1933	-8.4	-11.1	-5.4	-3.9	1.4	7.8	-8.3	-1.7	1.8	-0.9
1934	-4.7	-12.8	-3.8	-1.9	2.0	3.8	-7.1	-0.6	1.3	-1.4
1935	-5.0	-11.0	-5.5	-1.9	1.2	6.0	-7.2	-0.6	1.7	-0.9
1936	-6.4	-9.6	-4.4	-4.5	1.8	9.4	-6.8	-0.2	2.2	-0.5
1937	-7.0	-10.9	-5.7	-4.2	0.8	7.6	-7.9	-1.3	1.4	-1.3
1938	-7.0	-7.6	-6.1	-3.1	4.1	10.0	-6.9	-0.3	3.7	1.0
1939	-5.5	-6.9	-6.6	-4.8	0.0	7.0	-6.3	0.2	0.7	-2.0
1940	-3.0	-6.0	-4.3	-3.7	2.7	7.3	-4.4	2.1	2.1	-0.6
1941	-4.7	-7.9	-3.2	0.8	4.2	9.3	-5.2	1.3	4.8	2.1
1942	-5.1	-9.2	-7.8	-0.6	3.6	7.7	-7.4	-0.8	3.6	0.9
1943	-6.4	-9.1	-6.1	-2.9	1.6	5.3	-7.2	-0.6	1.3	-1.3
1944	-3.5	-6.8	-5.6	0.7	3.2	8.7	-5.3	1.3	4.2	1.5
1945	-9.0	-9.8	-7.5	-3.1	1.9	5.7	-8.8	-2.2	1.5	-1.2
1946	-6.0	-7.7	-3.8	-1.0	6.3	10.1	-5.8	0.8	5.1	2.5
1947	-9.0	-6.6	-5.7	0.8	4.4	7.9	-7.1	-0.6	4.3	1.7
1948	-5.1	-5.3	-4.6	-2.9	3.2	8.9	-5.0	1.6	3.1	0.4
1949	-7.5	-10.4	-6.1	-2.5	4.0	8.1	-8.0	-1.4	3.2	0.5
1950	-5.1	-9.2	-9.1	-2.4	0.6	7.5	-7.8	-1.3	1.9	-0.8
1951	-7.8	-8.1	-8.5	-3.3	2.1	7.0	-8.1	-1.6	1.9	-0.8
1952	-3.3	-6.9	-4.1	-2.1	4.5	6.9	-4.7	1.8	3.1	0.4

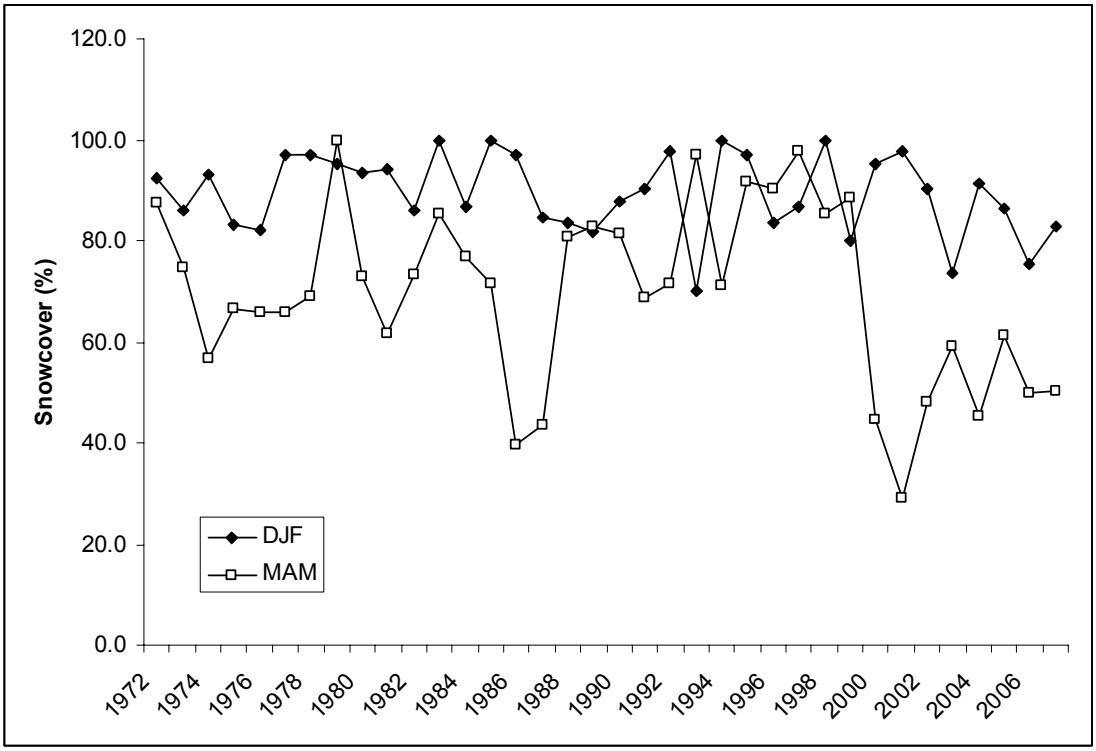
1953	-5.9	-6.5	-2.8	-1.4	3.2	7.4	-5.0	1.5	3.1	0.4
1954	-4.8	-7.9	-6.9	-2.8	3.4	8.1	-6.5	0.0	2.9	0.2
1955	-6.3	-6.8	-2.6	-0.5	2.0	5.2	-5.3	1.3	2.2	-0.5
1956	-3.8	-7.0	-5.3	-2.1	3.9	10.2	-5.4	1.2	4.0	1.3
1957	-6.3	-10.3	-8.5	-1.9	0.5	5.5	-8.4	-1.8	1.4	-1.3
1958	-5.1	-5.0	-4.4	0.3	4.6	5.4	-4.9	1.7	3.4	0.7
1959	-4.0	-7.9	-8.3	-1.6	4.6	6.3	-6.7	-0.2	3.1	0.4
1960	-8.1	-7.2	-2.7	-4.6	0.3	5.8	-6.0	0.6	0.5	-2.2
1961	-6.2	-6.8	-7.8	-1.3	1.7	8.9	-6.9	-0.4	3.1	0.4
1962	-5.2	-7.1	-3.6	0.2	2.8	6.5	-5.3	1.3	3.2	0.5
1963	-6.4	-5.8	-2.0	-1.6	3.7	6.2	-4.7	1.9	2.8	0.1
1964	-5.6	-12.6	-7.2	-0.6	1.9	6.5	-8.5	-1.9	2.6	-0.1
1965	-8.5	-6.2	-6.1	-2.1	1.1	6.6	-6.9	-0.4	1.8	-0.8
1966	-6.9	-4.3	-1.4	-1.8	1.8	6.7	-4.2	2.4	2.2	-0.4
1967	-5.5	-9.1	-5.2	-2.2	2.0	5.0	-6.6	0.0	1.6	-1.1
1968	-5.5	-7.8	-7.2	-1.1	2.5	4.9	-6.8	-0.3	2.1	-0.6
1969	-5.7	-10.4	-8.9	0.5	2.1	6.1	-8.4	-1.8	2.9	0.2
1970	-3.1	-6.7	-3.5	-2.1	5.3	9.3	-4.5	2.1	4.2	1.5
1971	-5.8	-8.6	-5.1	0.1	4.5	9.0	-6.5	0.1	4.6	1.9
1972	-3.2	-8.8	-14.2	-4.3	2.9	5.3	-8.7	-2.2	1.3	-1.4
1973	-8.4	-12.1	-4.1	-2.0	3.9	7.4	-8.2	-1.6	3.1	0.4
1974	-6.2	-9.2	-10.2	-0.2	4.9	7.2	-8.6	-2.0	4.0	1.3
1975	-6.7	-7.9	-6.9	-3.5	2.4	6.1	-7.2	-0.6	1.7	-1.0
1976	-6.3	-5.6	-7.2	-4.1	2.7	7.5	-6.4	0.2	2.0	-0.6
1977	-6.7	-12.5	-6.6	2.3	5.1	7.7	-8.6	-2.0	5.0	2.3
1978	-4.5	-8.4	-6.3	-2.9	5.0	7.9	-6.4	0.2	3.4	0.7
1979	-2.9	-6.2	-5.1	-3.0	5.2	4.0	-4.7	1.8	2.1	-0.6
1980	-4.3	-8.2	-7.7	-2.9	5.4	8.4	-6.7	-0.2	3.6	0.9
1981	-3.7	-5.5	-4.7	-0.4	4.2	8.2	-4.6	1.9	4.0	1.3
1982	-3.8	-6.5	-8.6	-3.6	3.7	7.7	-6.3	0.3	2.6	-0.1
1983	-7.9	-7.4	-5.2	-4.2	2.6	8.1	-6.8	-0.2	2.2	-0.5
1984	-6.1	-8.1	-10.6	0.7	4.2	7.9	-8.3	-1.7	4.2	1.6
1985	-7.0	-7.5	-3.2	-2.0	4.6	8.2	-5.9	0.7	3.6	0.9
1986	-5.5	-6.9	-5.7	-3.6	2.8	7.6	-6.0	0.5	2.2	-0.4
1987	-6.2	-5.5	-4.4	0.4	2.8	7.0	-5.4	1.2	3.4	0.7
1988	-2.2	-6.4	-6.1	-1.7	4.9	9.2	-4.9	1.7	4.1	1.4
1989	-2.9	-9.0	-8.4	-1.5	2.0	5.4	-6.8	-0.2	2.0	-0.7
1990	-3.2	-6.1	-6.0	-1.7	2.4	10.2	-5.1	1.5	3.6	0.9
1991	-6.2	-8.1	-7.2	-2.5	2.9	5.3	-7.2	-0.6	1.9	-0.8
1992	-3.7	-6.9	-6.0	-3.7	2.2	4.4	-5.5	1.0	1.0	-1.7
1993	-3.4	-8.3	-4.0	-3.0	4.2	7.0	-5.2	1.3	2.7	0.0
1994	-4.6	-6.0	-7.1	0.1	1.9	8.1	-5.9	0.7	3.4	0.7
1995	-5.4	-6.2	-4.8	-2.5	2.9	7.7	-5.5	1.1	2.7	0.0
1996	-6.1	-8.4	-4.9	-1.7	3.6	6.2	-6.5	0.1	2.7	0.0
1997	-4.7	-6.6	-6.5	-1.6	2.8	5.7	-5.9	0.6	2.3	-0.4
1998	-6.8	-8.7	-6.5	-2.7	4.6	6.9	-7.3	-0.8	2.9	0.2
1999	-3.6	-7.5	-2.9	-1.5	4.4	8.3	-4.7	1.9	3.7	1.0
2000	-3.6	-6.4	-6.4	-1.2	7.3	11.1	-5.5	1.1	5.7	3.0
2001	-3.9	-7.7	-4.6	-0.1	5.5	11.5	-5.4	1.2	5.6	3.0
2002	-3.8	-5.5	-5.4	-0.9	3.7	8.0	-4.9	1.7	3.6	0.9
MEAN	**-5.6**	**-7.9**	**-6.2**	**-2.1**	**3.0**	**7.1**	**-6.6**		**2.7**	

CRU precipitation dataset (mm) constructed from mean values of grids 495, 250 /498, 251 /500, 251

	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEPT	OCT	NOV	DEC	DJF	MAM	JJA	SON	ANNUAL
1902	200.7	221.7	710.3	658.0	203.0	95.3	24.3	2.0	29.3	528.0	568.7	141.0	187.8	523.8	40.6	375.3	3382.3
1903	648.7	706.0	987.0	595.7	1067.7	109.3	79.7	26.3	239.7	2.3	113.3	130.0	494.9	883.4	71.8	118.4	4705.7
1904	1102.0	104.3	1433.3	552.3	766.7	0.7	56.7	29.3	70.0	698.7	137.0	313.3	506.6	917.4	28.9	301.9	5264.3
1905	715.3	436.0	1104.7	741.3	472.0	48.0	6.7	157.7	102.0	10.3	33.7	280.0	477.1	772.7	70.8	48.7	4107.7
1906	147.0	1362.0	569.3	662.0	304.0	199.0	98.0	40.3	44.3	142.7	70.7	903.3	804.1	511.8	112.4	85.9	4542.7
1907	829.0	884.7	865.7	1749.3	969.3	492.3	51.0	32.3	8.3	516.0	132.3	696.3	803.3	1194.8	191.9	218.9	7226.7
1908	1334.7	689.0	533.3	1301.0	273.0	114.0	9.0	271.0	259.0	46.3	8.7	58.7	694.1	702.4	131.3	104.7	4897.7
1909	597.7	616.3	616.3	621.0	844.0	102.3	148.7	13.0	37.3	36.0	59.7	436.0	550.0	693.8	88.0	44.3	4128.3
1910	1489.3	305.7	1173.3	709.0	350.3	101.3	183.7	186.7	10.7	7.7	27.7	390.3	728.4	744.2	157.2	15.3	4935.7
1911	1238.3	660.3	1829.3	520.0	518.0	23.0	12.3	3.3	13.3	123.7	159.3	210.3	703.0	955.8	12.9	98.8	5311.3
1912	924.7	622.3	1037.3	1002.0	281.7	152.7	8.7	53.3	15.0	188.7	48.0	239.7	595.6	773.7	71.6	83.9	4574.0
1913	310.0	429.3	243.3	545.3	364.7	63.7	19.0	35.7	55.7	437.0	255.7	496.0	411.8	384.4	39.4	249.4	3255.3
1914	371.7	886.0	946.0	1014.7	264.0	258.7	117.7	30.7	114.0	814.3	729.3	935.0	730.9	741.6	135.7	552.6	6482.0
1915	6.0	1274.3	630.3	1916.3	112.0	110.0	29.3	11.3	40.3	62.7	15.7	338.0	539.4	886.2	50.2	39.6	4546.3
1916	310.0	409.0	894.7	960.3	313.7	111.7	19.7	275.3	47.3	9.3	3.0	100.7	273.2	722.9	135.6	19.9	3454.7
1917	402.7	241.3	553.3	25.3	121.7	41.0	3.7	140.7	174.0	43.0	3.3	44.0	229.3	233.4	61.8	73.4	1794.0
1918	251.3	561.0	1267.3	822.3	213.7	217.7	49.3	23.0	60.0	61.0	68.7	378.3	396.9	767.8	96.7	63.2	3973.7
1919	930.3	539.3	575.0	501.0	290.0	22.3	26.3	156.0	28.0	35.7	40.0	431.3	633.7	455.3	68.2	34.6	3575.3
1920	1009.7	614.0	1046.7	593.7	250.7	152.7	16.0	171.0	109.3	104.3	137.7	1558.7	1060.8	630.3	113.2	117.1	5764.3
1921	352.0	281.7	526.0	452.7	60.7	77.3	60.3	71.7	0.0	148.3	222.7	101.7	245.1	346.4	69.8	123.7	2355.0
1922	457.0	1298.0	699.3	253.0	534.3	165.7	85.0	96.3	128.7	167.7	71.3	1523.0	1092.7	495.6	115.7	122.6	5479.3
1923	576.0	1121.7	909.7	684.3	494.0	99.7	15.3	168.7	63.3	539.0	42.3	298.7	665.4	696.0	94.6	214.9	5012.7
1924	384.3	1247.0	720.3	1202.3	1101.3	84.7	88.7	12.3	225.7	67.3	179.3	127.7	586.3	1008.0	61.9	157.4	5441.0
1925	542.0	569.0	345.7	393.3	617.7	5.0	126.7	53.0	143.0	746.7	342.3	762.0	624.3	452.2	61.6	410.7	4646.3
1926	235.3	760.7	1238.7	1167.0	253.0	494.7	70.0	107.7	163.7	27.3	67.3	25.0	340.3	886.2	224.1	86.1	4610.3
1927	328.3	362.3	859.3	664.0	195.7	11.0	19.0	28.7	90.7	9.3	211.7	281.0	323.9	573.0	19.6	103.9	3061.0
1928	445.7	1223.7	1005.7	383.7	127.3	105.3	30.3	243.0	36.7	3.3	367.0	866.0	845.1	505.6	126.2	135.7	4837.7
1929	511.3	941.0	685.7	374.7	50.7	67.0	382.7	131.7	68.0	44.3	62.3	1035.3	829.2	370.3	193.8	58.2	4354.7
1930	700.3	765.7	622.7	1437.7	336.3	68.7	196.0	9.3	132.0	137.0	35.3	552.0	672.7	798.9	91.3	101.4	4993.0
1931	1030.7	636.0	649.7	763.0	1262.0	97.0	189.0	52.0	34.3	396.3	707.3	286.7	651.1	891.6	112.7	379.3	6104.0
1932	1279.3	398.7	1110.7	221.3	310.3	69.3	54.3	110.0	22.7	191.7	8.3	807.0	828.3	547.4	77.9	74.2	4583.7
1933	489.7	631.3	1394.3	634.7	290.7	204.7	22.0	127.7	81.0	18.7	127.0	314.0	478.3	773.2	118.1	75.6	4335.7
1934	486.7	745.0	867.0	1006.7	725.7	77.0	0.7	13.7	81.0	123.3	85.0	841.7	691.1	866.4	30.4	96.4	5053.3
1935	1187.7	1017.3	563.0	988.7	218.0	542.0	36.0	185.7	79.0	177.3	559.0	754.7	986.6	589.9	254.6	271.8	6308.3
1936	218.3	1143.3	958.3	927.3	199.7	63.3	6.7	44.7	31.0	230.0	599.7	74.0	478.6	695.1	38.2	286.9	4496.3
1937	452.3	1025.3	722.7	1302.3	275.3	6.3	49.3	4.0	50.3	420.7	101.7	652.3	710.0	766.8	19.9	190.9	5062.7
1938	892.0	659.7	1120.7	428.7	80.0	114.7	40.7	35.0	24.3	13.7	238.3	456.0	669.2	543.1	63.4	92.1	4103.7
1939	539.3	1331.3	1835.7	539.0	236.7	0.7	30.3	11.0	3.7	291.0	76.7	447.3	772.7	870.4	14.0	123.8	5342.7
1940	871.7	419.0	1306.3	493.7	191.0	193.3	15.7	39.7	212.7	327.7	136.7	168.3	486.3	663.7	82.9	225.7	4375.7
1941	514.7	436.0	675.3	898.0	210.7	33.0	38.3	57.0	28.3	58.0	118.0	468.0	472.9	594.7	42.8	68.1	3535.3
1942	1252.0	1257.3	464.0	752.7	316.7	164.7	135.0	24.7	59.7	212.7	327.7	727.0	1078.8	511.1	108.1	200.0	5694.0
1943	930.3	270.7	971.7	917.3	449.7	42.3	59.0	119.7	33.0	534.3	265.7	212.7	471.2	779.6	73.7	277.7	4806.3
1944	566.7	1151.0	901.0	1125.0	213.7	16.3	85.0	141.3	67.0	176.7	205.7	239.0	652.2	746.6	80.9	149.8	4888.3
1945	856.7	496.0	1074.0	853.0	520.7	340.7	38.3	72.7	52.0	46.7	665.0	180.3	511.0	815.9	150.6	254.6	5196.0
1946	687.3	1053.0	537.0	127.0	78.0	100.3	102.0	31.3	18.0	216.0	201.0	246.7	662.3	247.3	77.9	145.0	3397.7
1947	527.7	1091.7	737.3	188.7	145.0	63.7	88.0	52.7	75.7	107.3	387.3	316.0	645.1	357.0	68.1	190.1	3781.0
1948	218.7	494.7	940.7	963.3	208.3	94.7	22.7	56.0	139.7	185.3	81.3	254.3	322.6	704.1	57.8	135.4	3659.7
1949	969.7	735.7	1023.0	514.7	157.7	562.7	43.0	64.3	90.0	125.7	77.3	1035.3	913.6	565.1	223.3	97.7	5399.0
1950	1073.3	345.0	705.0	779.3	516.0	61.0	48.0	192.7	14.3	43.0	224.0	378.0	598.8	666.8	100.6	93.8	4379.7
1951	874.3	693.7	1136.3	577.7	846.0	87.3	127.0	36.0	12.7	507.3	236.7	41.7	536.6	853.3	83.4	252.2	5176.7
1952	657.7	1266.7	957.7	955.7	347.3	114.0	90.7	146.3	19.0	34.7	73.3	392.7	772.3	753.6	117.0	42.3	5055.7
1953	524.0	911.3	1326.7	701.3	485.3	200.0	45.7	207.3	75.0	378.0	374.7	324.3	586.6	837.8	151.0	275.9	5553.7
1954	1546.3	1101.0	1007.3	570.3	355.3	146.0	190.3	57.3	24.0	122.3	262.0	662.7	1103.3	644.3	131.2	136.1	6045.0
1955	212.0	303.3	1485.7	277.0	925.3	163.3	152.7	64.7	71.3	39.3	78.7	284.3	266.6	896.0	126.9	63.1	4057.7

1956	540.3	1016.7	1710.7	1111.7	105.0	239.7	107.7	72.0	40.0	427.0	350.0	549.0	702.0	975.8	139.8	272.3	6269.7
1957	1179.0	755.3	1323.0	1190.3	524.3	200.7	99.7	5.0	10.0	560.3	787.7	285.7	740.0	1012.6	101.8	452.7	6921.0
1958	778.7	395.7	1074.3	1121.3	581.0	129.0	59.3	115.7	85.3	104.0	285.3	987.3	720.6	925.6	101.3	158.2	5717.0
1959	629.0	823.3	737.0	667.3	900.7	47.0	39.0	36.7	58.7	21.3	878.3	1070.7	841.0	768.3	40.9	319.4	5909.0
1960	926.7	810.0	1164.3	1327.3	695.0	76.0	81.0	51.3	176.7	46.3	692.7	737.0	824.6	1062.2	69.4	305.2	6784.3
1961	559.7	314.7	1143.3	1879.3	122.3	126.7	42.7	352.7	32.3	157.3	345.3	388.3	420.9	1048.3	174.0	178.3	5464.7
1962	196.0	532.7	585.7	1088.0	817.3	96.7	80.7	102.0	91.0	150.0	343.0	371.3	366.7	830.3	93.1	194.7	4454.3
1963	65.0	605.7	1024.0	1167.7	1556.7	86.0	83.7	48.7	76.3	204.7	532.3	399.3	356.7	1249.4	72.8	271.1	5850.0
1964	901.7	1268.7	890.7	1382.0	563.3	105.3	260.7	5.3	19.3	49.7	315.3	285.3	818.6	945.3	123.8	128.1	6047.3
1965	947.3	1079.3	729.3	1410.7	546.7	19.3	251.0	27.7	62.7	72.0	572.0	297.3	774.7	895.6	99.3	235.6	6015.3
1966	240.0	1103.3	1547.0	817.3	304.3	2.0	60.7	101.7	117.7	396.3	45.3	164.0	502.4	889.6	54.8	186.4	4899.7
1967	164.0	1013.3	773.0	1368.7	477.7	337.7	121.7	23.3	89.7	579.7	210.0	256.3	477.9	873.1	160.9	293.1	5415.0
1968	626.0	429.0	1030.0	1082.3	599.7	180.0	15.0	25.7	1.3	256.7	232.3	620.7	558.6	904.0	73.6	163.4	5098.7
1969	809.3	656.3	1215.7	1020.3	708.7	303.3	240.7	10.7	33.7	379.3	580.7	1194.0	886.6	981.6	184.9	331.2	7152.7
1970	571.0	283.0	725.7	585.3	181.7	33.0	224.7	21.7	83.0	120.3	71.7	136.0	330.0	497.6	93.1	91.7	3037.0
1971	161.0	768.7	715.3	549.0	157.7	20.7	26.7	19.3	27.7	96.0	151.3	323.7	417.8	474.0	22.2	91.7	3017.0
1972	1038.7	813.3	1342.7	876.3	1156.3	328.3	121.7	15.7	40.7	184.7	300.0	227.3	693.1	1125.1	155.2	175.1	6445.7
1973	475.3	1144.0	1700.0	812.0	286.7	24.7	62.3	24.0	194.3	68.7	87.7	635.0	751.4	932.9	37.0	116.9	5514.7
1974	468.3	902.3	524.7	677.7	448.0	54.7	121.3	21.7	55.3	9.0	125.3	207.7	526.1	550.1	65.9	63.2	3616.0
1975	464.7	844.0	989.7	1154.0	632.0	65.7	20.3	63.7	8.0	165.0	187.0	685.3	664.7	925.2	49.9	120.0	5279.3
1976	580.3	1218.0	921.0	738.7	306.0	51.3	20.3	32.3	24.3	352.0	54.7	378.3	725.6	655.2	34.7	143.7	4677.3
1977	1234.3	98.3	249.7	436.7	311.3	67.3	59.0	8.0	24.7	338.0	472.0	188.0	506.9	332.6	44.8	278.2	3487.3
1978	739.7	539.3	983.0	649.3	474.7	36.7	88.3	8.0	0.0	86.0	627.0	526.7	601.9	702.3	44.3	237.7	4758.7
1979	489.0	672.3	829.0	699.3	809.7	50.0	6.0	120.7	73.0	68.3	149.3	489.0	550.1	779.3	58.9	96.9	4455.7
1980	875.3	1116.0	1369.0	592.7	238.7	173.3	62.7	57.7	12.7	450.0	224.7	402.3	797.9	733.4	97.9	229.1	5575.0
1981	945.7	1286.3	914.3	756.7	478.7	53.3	234.0	78.7	41.3	355.3	249.7	175.7	802.6	716.6	122.0	215.4	5569.7
1982	444.0	757.0	1319.3	87.7	429.7	49.0	153.7	4.0	135.7	410.3	914.3	501.0	567.3	612.2	68.9	486.8	5205.7
1983	422.7	634.3	1158.7	728.7	567.0	237.7	136.0	71.7	145.7	139.7	254.3	482.7	513.2	818.1	148.4	179.9	4979.0
1984	220.0	616.3	1289.0	803.7	244.7	70.0	76.3	29.0	52.3	67.0	516.0	335.3	390.6	779.1	58.4	211.8	4319.7
1985	750.7	242.7	575.0	803.7	382.3	61.3	12.0	197.7	6.0	101.7	52.7	707.0	566.8	587.0	90.3	53.4	3892.7
1986	230.3	449.3	1131.0	464.3	661.7	102.7	75.7	178.0	45.0	92.3	531.3	569.0	416.2	752.3	118.8	222.9	4530.7
1987	119.0	454.0	1886.0	1174.3	669.0	300.3	265.7	44.3	111.0	900.0	116.3	961.0	511.3	1243.1	203.4	375.8	7001.0
1988	705.7	573.7	1643.0	512.7	287.7	84.0	30.3	298.3	4.0	46.3	3.3	232.3	503.9	814.4	137.6	17.9	4421.3
1989	688.0	551.0	1089.3	526.3	480.3	351.7	327.0	420.7	184.3	38.7	380.0	900.0	713.0	698.7	366.4	201.0	5937.3
1990	1067.0	734.7	606.0	953.7	200.7	223.0	23.3	590.0	11.3	66.0	55.3	633.0	811.6	586.8	278.8	44.2	5164.0
1991	1106.7	837.3	1743.0	589.7	1245.3	161.0	108.7	10.3	458.3	31.0	230.0	815.3	919.8	1192.7	93.3	239.8	7336.7
1992	786.7	722.0	896.3	1306.3	614.7	186.0	62.7	51.3	50.7	347.7	237.3	685.0	731.2	939.1	100.0	211.9	5946.7
1993	258.7	515.0	1293.3	674.3	858.0	70.0	29.3	23.3	14.0	249.7	472.7	308.0	360.6	941.9	40.9	245.4	4766.3
1994	749.7	455.7	501.7	456.7	641.7	77.7	121.0	83.0	99.3	974.0	173.7	792.7	666.0	533.3	93.9	415.7	5126.7
1995	166.3	1318.7	962.0	920.0	407.3	249.0	76.0	89.0	40.0	303.3	78.0	467.3	650.8	763.1	138.0	140.4	5077.0
1996	246.3	693.7	1446.7	781.0	685.3	216.0	83.0	75.0	175.0	414.0	192.7	297.3	412.4	971.0	124.7	260.6	5306.0
1997	274.3	335.3	1396.0	1248.7	886.0	143.0	64.3	29.0	25.7	851.0	235.3	40.0	216.6	1176.9	78.8	370.7	5528.7
1998	1068.0	1240.3	1029.3	1148.0	1223.3	108.3	189.7	125.0	141.0	192.3	64.7	322.3	876.9	1133.6	141.0	132.7	6852.3
1999	787.3	907.0	1130.0	379.7	390.7	134.0	41.7	38.7	132.3	138.3	973.3	124.7	606.3	633.4	71.4	414.7	5177.7
2000	582.0	465.3	501.0	138.7	71.0	187.3	50.7	44.7	99.7	174.7	63.3	23.7	357.0	236.9	94.2	112.6	2402.0
2001	285.0	341.3	537.7	299.0	155.7	203.7	101.3	217.7	49.0	178.7	341.0	460.0	362.1	330.8	174.2	189.6	3170.0
2002	337.7	665.3	650.0	816.3	442.3	280.7	116.0	104.7	3.7	68.0	243.7	533.7	512.2	636.2	167.1	105.1	4262.0
MEAN	**634.8**	**730.2**	**971.3**	**791.9**	**463.8**	**134.2**	**84.0**	**86.2**	**73.0**	**220.0**	**251.8**	**459.3**	**608.1**	**742.3**	**101.5**	**181.6**	**4900.4**

APPENDIX II: SNOW COVER DATA



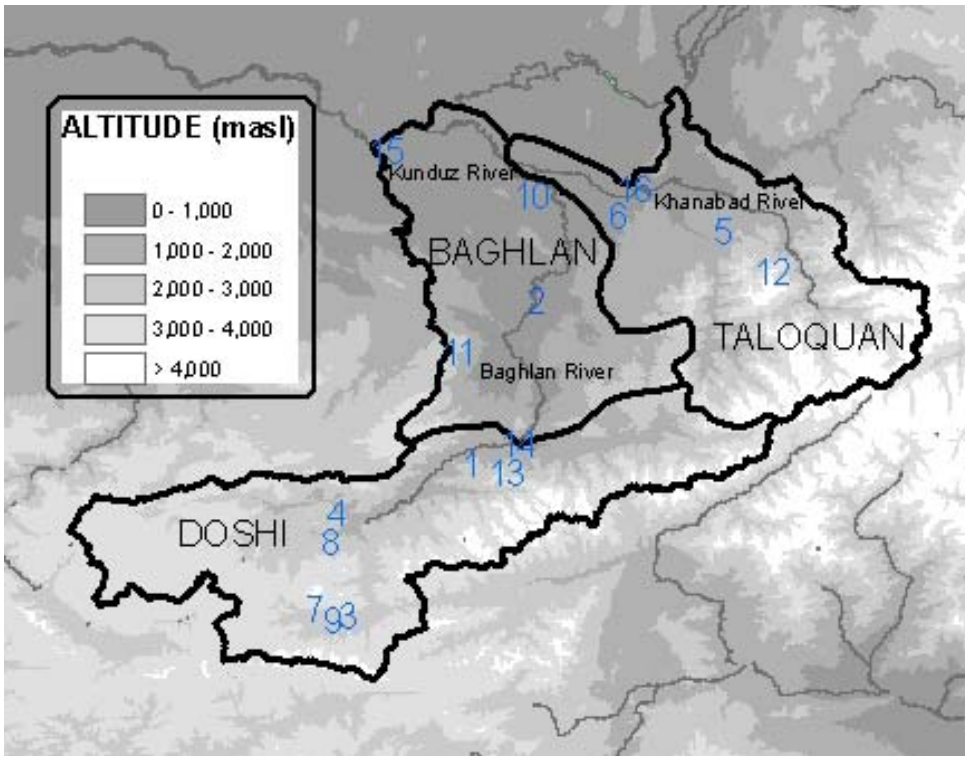
Snowcover (%) time series for winter DJF and spring (MAM) 1972-2007.

Constructed snow cover time series. % cover of Kunduz River Basin above 2500masl. Anomalies represent % deviation from 1972-07 mean.

YEAR	DEC	JAN	FEB	MAR	APR	MAY	SEASON	DJF MEAN	DJF ANOMALY	MAM MEAN	MAM ANOMALY
1972	92.6	95.4	88.9	100.0	88.9	73.3	89.8	92.3	2.9	87.4	18.1
1973	100.0	76.9	81.5	100.0	79.6	44.4	80.4	86.1	-3.3	74.7	5.4
1974	100.0	83.7	95.4	79.6	79.6	11.1	74.9	93.0	3.6	56.8	-12.5
1975	77.8	81.5	90.7	85.2	75.3	38.9	74.9	83.3	-6.1	66.5	-2.8
1976	68.5	96.3	81.5	88.9	70.4	38.5	74.0	82.1	-7.3	65.9	-3.4
1977	100.0	100.0	90.7	76.3	93.8	27.8	81.4	96.9	7.5	66.0	-3.3
1978	91.1	100.0	100.0	81.5	100.0	25.9	83.1	97.0	7.6	69.1	-0.2
1979	85.2	100.0	100.0	100.0	100.0	100.0	97.5	95.1	5.7	100.0	30.7
1980	80.0	100.0	100.0	100.0	100.0	18.5	83.1	93.3	3.9	72.8	3.5
1981	100.0	82.2	100.0	100.0	57.4	27.8	77.9	94.1	4.7	61.7	-7.6
1982	76.3	81.5	100.0	100.0	68.5	51.1	79.6	85.9	-3.5	73.2	3.9
1983	100.0	100.0	100.0	96.3	100.0	59.3	92.6	100.0	10.6	85.2	15.9
1984	60.0	100.0	100.0	91.1	90.7	48.1	81.7	86.7	-2.7	76.7	7.4
1985	100.0	100.0	100.0	100.0	59.3	55.6	85.8	100.0	10.6	71.6	2.3
1986	100.0	100.0	90.7	65.2	35.0	18.5	68.2	96.9	7.5	39.6	-29.7
1987	100.0	85.2	68.5	73.3	38.9	18.5	64.1	84.6	-4.8	43.6	-25.7
1988	51.1	100.0	100.0	100.0	85.2	57.4	82.3	83.7	-5.7	80.9	11.6
1989	63.7	90.7	90.7	85.2	100.0	63.7	82.3	81.7	-7.7	83.0	13.7
1990	92.6	70.4	100.0	100.0	90.7	53.3	84.5	87.7	-1.7	81.4	12.1
1991	82.2	88.9	100.0	100.0	63.9	42.2	79.5	90.4	1.0	68.7	-0.6
1992	92.6	100.0	100.0	100.0	77.8	37.0	84.6	97.5	8.1	71.6	2.3
1993	96.3	100.0	13.9	100.0	90.7	100.0	83.5	70.1	-19.3	96.9	27.6

1994	100.0	100.0	100.0	83.7	72.9	57.4	85.7	100.0	10.6	71.3	2.0
1995	91.1	100.0	100.0	100.0	100.0	74.8	94.3	97.0	7.6	91.6	22.3
1996	83.7	66.7	100.0	88.9	100.0	82.2	86.9	83.5	-5.9	90.4	21.1
1997	68.9	91.1	100.0	92.6	100.0	100.0	92.1	86.7	-2.7	97.5	28.2
1998	100.0	100.0	100.0	100.0	88.9	66.7	92.6	100.0	10.6	85.2	15.9
1999	40.0	100.0	100.0	100.0	100.0	65.9	84.3	80.0	-9.4	88.6	19.3
2000	85.2	100.0	100.0	77.8	51.9	3.7	69.8	95.1	5.7	44.4	-24.9
2001	92.6	100.0	100.0	54.1	18.5	14.8	63.3	97.5	8.1	29.1	-40.2
2002	88.0	89.5	93.3	71.8	50.9	21.5	69.2	90.3	0.9	48.1	-21.2
2003	75.3	60.9	85.1	85.2	58.7	33.0	66.4	73.8	-15.6	59.0	-10.3
2004	92.8	91.2	89.6	74.6	39.7	21.5	68.2	91.2	1.8	45.3	-24.0
2005	73.3	96.8	88.8	84.8	63.4	36.0	73.9	86.3	-3.1	61.4	-7.9
2006	68.1	88.4	69.6	74.9	57.3	17.5	62.6	75.4	-14.0	49.9	-19.4
2007	92.2	83.1	73.0	76.1	55.6	19.3	66.5	82.8	-6.6	50.3	-19.0
MEAN	**85.0**	**91.7**	**91.4**	**88.5**	**75.1**	**45.2**	**79.5**	**89.4**		**69.6**	

APPENDIX III: RIVER DISCHARGE DATA



Position of gauging stations within the Kunduz River Basin. All three sub-basins (Baghlan, Taloquan and Doshi) are well represented.

Gauging station information

Station Number	River	Station Name	Location		Distance from source	Elevation (masl)	Catchment Area Km2
			Lat	Long			
15	Kunduz	Kulukh-Tepa	36˚ 59´ N	68˚ 18´ E	568	320	37100
16	Khanabad	Pul-i-Chugha	36˚ 44 ´N	69 ˚12´ E	185	558	9760
6	Bangi	Pul-i-Bangi	36˚ 42´ N	65˚ 15´ E	142	610	4200
5	Farkhar	near Taloqan	36˚ 38´ N	69˚ 43´ E	130	991	4110
10	Baghlan	Char Dara	36˚ 42´ N	68˚ 50´ E	452	401	24820
2	Baghlan	Gerdab	36˚ 22´ N	68º 52´ E	380	464	22930
11	Baghlan	Baghlan	36˚ 06 ´ N	68˚ 40´ E	321	562	19740
14	Andarab	Doshi	35˚ 36´ N	68˚ 41´ E	111	860	3705
1	Surkhab	Pul-i-Konda sang	35˚ 36´N	68˚ 36´ E	241	899	12610
8	Bamyan	Doab	35˚ 16´ N	67˚ 59´ E	129	1468	5005
3	Bamyan	Ahangaran	34˚ 49´N	67˚ 55´ E	70	2450	1660
7	Bamyan	Bamyan	34˚ 50 N	67˚ 49´ E	61	2506	945
9	Foldari	Bamyan	34˚ 49´ N	67˚ 49´ N	31	2507	320
4	Surkhab	Dasht-i-Sefed	35˚ 18´ N	67˚ 55´ E	154	1588	3795
13	Baghlan	Pul-i-Khumri	35˚ 36´ N	68˚ 43´ E	293	639	17410
12	Farkhar	Pul-i-Mastan	36˚ 27´ N	69˚ 54´ E	97	1404	3190

June mean discharge anomalies (% deviation from station June mean) for 14 gauging stations (3,12 omitted as insufficient data points) used to construct overall basin discharge anomaly (BDA) time series 1965-80.

Year	1	2	4	5	6	7	8	9	10	11	13	14	15	16	Basin Discharge Anomaly
1965		13.0			2.7				13.0		11.0	22.2		3.2	10.9
1966		24.3			59.4				28.5		-1.3	37.5	31.8	-47.9	18.9
1967		9.2		-14.3	-10.7				14.0		2.1	21.4	90.3	-11.2	12.6
1968	74.4	89.1	53.7	19.9	59.4		66.7		117.3	99.8		91.0	49.1	37.4	68.9
1969	19.7	26.2	16.9	43.3	42.5		83.3		32.8	39.0		33.3	-39.0	52.4	31.9
1970	-36.0	-48.0	-28.5	-12.8	-26.9	-43.9	-46.4	-45.9	-48.5	-40.3		-48.8	-54.4	-13.9	-38.0
1971	-56.1	-63.4	-46.2	-15.1	-50.8	-48.5	-69.4	43.5	-61.3	-57.5		-51.0	-12.2	-36.4	-40.3
1972	0.7	12.5	59.6	-16.7	-22.1	-18.9	20.3	153.6	15.1	15.6		18.0	12.1	-20.9	17.6
1973	31.3	23.9	-19.1	-18.2	-5.9	214.4	42.8	37.8	29.0	25.7		10.4	-44.3	-8.0	24.6
1974	-21.4	-45.2	-35.7	-33.8	-37.8	-60.6	-29.7	-32.8	-51.3	-40.9		-46.7	-6.4	-26.7	-36.1
1975	-3.5	-3.5		-12.0	-1.7	-13.6	-17.0	12.7	0.6	-2.0		-14.3	-35.0	13.9	-6.3
1976	-7.7	-28.1		-24.8	-29.8	-26.5	-18.0	-40.2	-33.1	-21.2		-31.1	-3.3	28.3	-19.6
1977	-13.3	-13.5		18.4	12.0		-21.9		-11.7	-9.4		-17.1	-11.9	7.5	-6.1
1978	-18.4	-43.3		25.4	33.3		-7.5		-48.5	-33.4		-31.8	20.4		-11.5
1979	30.3	10.2		42.5	17.1				4.4	24.6		7.8			19.6
1980		43.7		-1.9	-40.8										0.4